Portable Polarization: Partisan Identity and Health Behavior During COVID-19

2026-03-12

Abstract

COVID-19 mortality diverged along partisan lines, yet whether individual partisans genuinely differed in health behavior remains contested. Using data from the Berkeley Interpersonal Contact Study (18,233 respondents, six waves, April 2020-May 2021), we test competing explanations for partisan gaps in social contacts and mask-wearing. Partisan differences persist within shared policy environments, across regions, and after demographic adjustment – a pattern we term "portable polarization." The two behaviors diverge in revealing ways: rising mortality temporarily compressed the contact gap, a private, high-cost behavior, but left mask-wearing polarization untouched, consistent with visible behaviors functioning as identity markers rather than responses to risk. Stated concern fails to account for behavioral differences, suggesting that expressed attitudes on politicized topics may not reliably predict protective behavior. Partisan identity is independently associated with health behavior through channels not reducible to perception, policy, or place.

Manuscript word count: 8,483 (text: 7,283; references est.: 1,200) | Tables: 1 | Figures: 5 |

Running head: Partisan Differences in Health Behaviors

The COVID-19 pandemic revealed a striking pattern in American public health: mortality rates diverged sharply along partisan lines. Counties that voted heavily for Donald Trump in 2020 experienced substantially higher COVID-19 death rates than those that supported Joe Biden (Gollwitzer et al. 2020; Wallace, Goldsmith-Pinkham, and Schwartz 2023). This partisan mortality divide was not a statistical artifact; it emerged during the pandemic and widened over time, particularly after vaccines became available (Wallace et al. 2023). Early in the pandemic, mortality was higher in

Democratic-leaning counties, reflecting the initial concentration of cases in urban areas like New York City; but by fall 2020, this pattern reversed, and Republican-leaning counties experienced persistently higher mortality through the remainder of the pandemic , see 1.

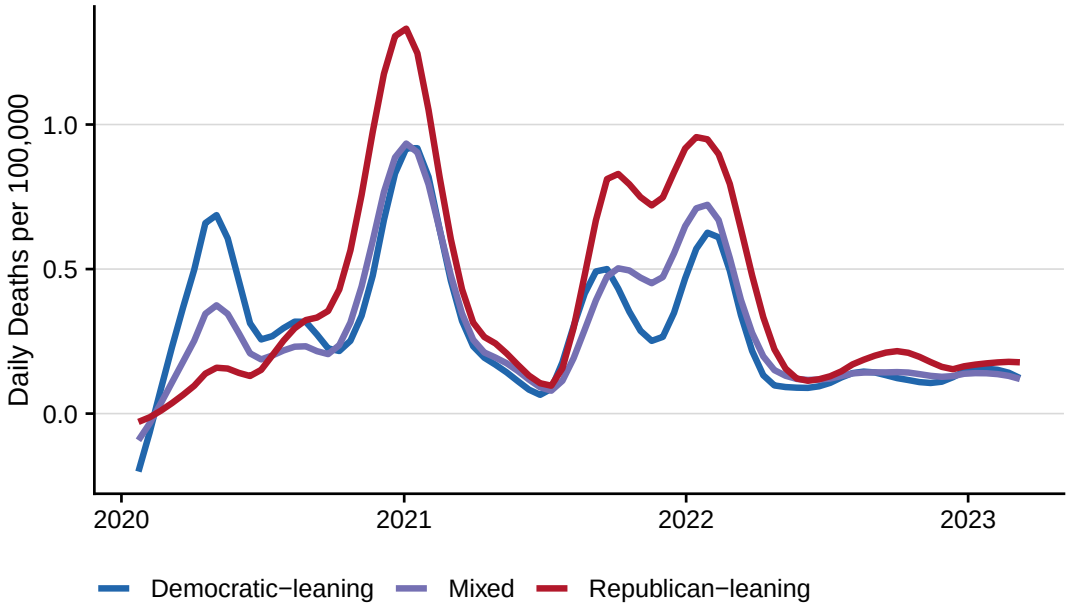

Figure 1: COVID-19 Mortality by County Partisan Composition

These aggregate patterns raise the possibility that individual partisans differed in their behavioral responses to the pandemic. But county-level data alone cannot tell us whether such individual-level differences exist, how large they are, or whether they persist once we account for the many other ways that Republican- and Democratic-leaning places differ. Survey research has documented substantial partisan gaps in pandemic-related attitudes, with Republicans expressing less concern about the virus and lower support for mitigation measures (Gadarian, Goodman, and Pepinsky 2021), and some studies have linked these attitudinal differences to downstream health consequences, finding that political orientation predicted self-rated health and well-being even after accounting for local pandemic severity (Coleman and Andersson 2024). Yet this literature leaves critical questions unanswered: attitudinal studies capture what partisans *say* but not necessarily what they *do*, and

analyses that link partisanship to health outcomes cannot isolate whether behavioral divergence or geographic sorting drives the association. Some studies have begun to examine contextual moderation, notably Baxter-King et al. (2022)'s finding that local partisan composition shaped mask-wearing asymmetrically across parties, but existing work has not tested multiple behaviors within the same sample across comparable policy, geographic, and demographic contexts. Before we can explain *why* partisan identity shapes health behavior, we must first establish *whether* it does so robustly across these dimensions, and *under what conditions* partisan identity matters most.

This paper uses data from the Berkeley Interpersonal Contact Study (BICS), a nationally representative survey that collected detailed counts of daily interpersonal contacts and mask usage during the COVID-19 pandemic, to adjudicate between competing explanations for partisan health gaps. Rather than testing a single mechanism, we exploit variation across policy environments, demographic composition, geographic context, perceived severity, and pandemic conditions to ask: which theories survive contact with the empirical patterns? If partisan differences are artifacts of policy or place, they should dissolve within comparable environments; if they reflect demographic sorting, they should close after controlling for covariates; if perception fully drives behavior, they should disappear among partisans who assess risk similarly; and if objective severity overrides identity, they should narrow when mortality is highest.

Our findings reveal that partisan differences in health behaviors are remarkably robust. Gaps in social contacts and mask-wearing persist within policy environments, across regions, in both urban and rural areas, and across diverse local partisan compositions, a pattern we term "portable polarization." At the same time, the magnitude of polarization varies in theoretically revealing ways: objective severity temporarily compresses contact gaps but leaves mask-wearing polarization un-

touched, and stated concern fails to account for behavioral differences, suggesting that expressed attitudes on politicized health topics may function as markers of group allegiance rather than guides to behavior. These findings advance sociological understanding of how partisan identity, increasingly a master status in American life (Mason 2018), shapes health behavior through channels that operate independently of perception, policy, and place.

1 Background

1.1 Partisan Differences in Health Behavior

A growing literature documents partisan differences in health-related attitudes and behaviors during the COVID-19 pandemic. In one of the earliest and most influential studies, Gadarian et al. (2021) showed that partisan identity predicted attitudes toward social distancing, mask-wearing, and government intervention even after controlling for demographics, local case rates, and media consumption, suggesting that party identification operated as more than a proxy for other characteristics (see also Allcott et al. 2020; Clinton et al. 2021; Grossman et al. 2020). These attitudinal differences appeared to have real consequences: county-level analyses linked Republican vote share to higher COVID-19 case and death rates, particularly after the initial wave (Gollwitzer et al. 2020; Krieger et al. 2022; Neelon et al. 2021), and Coleman and Andersson (2024) found that political orientation predicted individual self-rated health and well-being during the pandemic even after accounting for local severity. However, most of this work relies on aggregate data or attitudinal measures rather than direct assessments of individual behavior, leaving open the question of whether individual partisans actually *behaved* differently or whether observed patterns reflect the

demographic and geographic sorting of partisans into different environments.

Studies that do examine individual behavior generally confirm partisan differences, but with important limitations. Milosh et al. (2021) found that mask adoption was 12 percentage points lower in Republican-leaning counties even after accounting for local case rates, and Kaushal et al. (2022) documented persistent partisan gaps in social distancing that widened over time (see also Pink et al. 2021). County-level GPS data told a similar story: Allcott et al. (2020) found that Republicans engaged in significantly less social distancing even after controlling for local disease severity (see also Gollwitzer et al. 2020). Clinton et al. (2021), using over one million individual survey responses, found that partisan differences in social activity were initially small but widened substantially over time, with Republicans increasing social activities by 75% compared to 47% for Democrats, and that these gaps far exceeded differences between the most and least severely affected communities. Yet even these studies leave key questions unresolved: the county-level analyses cannot separate individual identity from place-level confounds, and the individual-level survey work, while documenting the temporal arc, did not systematically test whether partisan gaps survive within shared policy, geographic, and demographic contexts. A Democrat in San Francisco and a Republican in rural Alabama differ not only in party identification but in the mandates they face, the norms that surround them, and the information they receive. Existing work has not systematically tested whether partisan behavioral gaps survive within shared contexts.

This study builds on this body of work by moving from aggregate patterns and attitudinal measures to individual-level behavioral data: actual counts of daily contacts and mask usage during specific interactions. Crucially, the design of our data allows us to test whether partisan differences persist within comparable policy, geographic, and demographic contexts, the comparison that existing

work has largely been unable to make. By holding context constant and asking whether partisan gaps survive, we can begin to distinguish identity-driven behavioral divergence from the confounding effects of place.

1.2 Partisanship as Social Identity and the Politicization of Health

The transformation of partisanship from a set of policy preferences into a fundamental social identity provides the theoretical foundation for expecting behavioral divergence. Over the past several decades, party identification in the United States has evolved into a primary group attachment that rivals race, religion, and ethnicity in psychological salience (Green et al. 2020; Iyengar and Westwood 2015; Kalmoe and Mason 2022). Americans increasingly select friends, neighbors, and romantic partners based on party affiliation (Mason 2018), and even sort into geographically distinct communities along partisan lines (Brown and Enos 2021); they express greater social distance from and animosity toward members of the opposing party; and they view events through the lens of partisan group membership rather than ideological principle (Iyengar and Krupenkin 2018; Iyengar and Westwood 2015). Social identities do not merely correlate with behavior; they drive it, through mechanisms including in-group conformity, social pressure, and responsiveness to elite cues (Hoog and Pat-El 2024). Health sociology has increasingly recognized this dynamic: research on health lifestyles shows that health behaviors cluster into integrated patterns undergirded by group-based identities and norms rather than individual calculation (Cockerham 2005; Mollborn, Lawrence, and Onge 2021), suggesting that the same identity-driven processes documented in partisan politics may operate in the health domain.

These dynamics found fertile ground in the domain of public health. Trust in scientific institutions

had already been diverging along ideological lines well before COVID-19: Gauchat (2012) documented a widening gap in trust in science among conservatives over the period from the 1970s to 2010, a decline concentrated among the most educated, suggesting that the politicization of science is not a knowledge deficit but an identity-driven process. Parallel data on partisan identification tell a similar story: confidence in science among Republicans dropped sharply beginning in the mid-2000s, even as Democratic confidence held steady or increased (Jones 2021). The pandemic dramatically accelerated this trend (Van Bavel et al. 2024), and recent work in this journal has shown that the resulting polarization is not simply a matter of knowledge deficits: Gauchat (2025) found that conservative distrust of scientific authorities on vaccination was actually *magnified* among the scientifically literate. Questions that epidemiologists viewed as empirical (How severe is the virus? Do masks reduce transmission? Are vaccines safe and effective?) became markers of partisan allegiance (Gadarian et al. 2021; Rothgerber et al. 2020). Republican political leaders and conservative media figures frequently downplayed the pandemic's severity and questioned the efficacy of protective behaviors, while Democratic leaders emphasized the threat and urged compliance with public health guidance (Gadarian, Goodman, and Pepinsky 2024; Green et al. 2020).

If partisanship functions as a social identity, and if health behaviors became symbols of that identity, we should expect partisans to have diverged in their behavioral responses, not merely because they lived in different policy environments, but because their social identities motivated different responses to the same threat. The question is not whether partisans expressed different attitudes about the pandemic (they clearly did) but whether they *reported* different behaviors, and whether those behavioral differences can be reduced to the contexts in which partisans are embedded or

whether they reflect the independent force of identity itself.

1.3 Competing Explanations for Partisan Health Gaps

We organize our analysis around a set of competing explanations for partisan behavioral gaps, each of which generates testable predictions about when and where gaps should appear or disappear. Our theoretical expectation, grounded in social identity theory (Tajfel and Turner 1979), health lifestyles research (Cockerham 2005), and symbolic boundary theory (Lamont 2000), is that partisan identity directly drives health behavior; the competing explanations represent alternative accounts that could produce the same observed patterns without invoking identity as a causal force. This approach, using patterns of contextual variation to adjudicate between explanations, allows us to constrain the set of plausible accounts even though we cannot directly test causal mechanisms. The framework yields two core insights: (1) each alternative explanation implies that partisan gaps should narrow or vanish under specific conditions, so persistent gaps under those conditions constitute evidence against that explanation; and (2) if gaps persist across *all* conditions where alternative explanations predict they should disappear, the surviving account is that partisan identity itself drives behavior.

The most straightforward alternative explanations emphasize policy environments and demographic composition. Democrats were concentrated in urban areas governed by officials who implemented stricter lockdowns, mask mandates, and business closures; Republicans lived in areas with fewer restrictions. If partisan gaps are artifacts of these policy differences, they should disappear when we compare partisans within the same policy regime: within strict mandate areas, within lenient areas, and within areas with no mandates. Similarly, if gaps reflect the fact

that Democrats and Republicans differ in age, education, urbanicity, and other characteristics associated with health behavior, controlling for these demographics should eliminate the gaps.

A deeper challenge comes from explanations rooted in local context and social conformity. Research has documented powerful contextual effects on health outcomes. For example, Montez et al. (2025) showed that county-level characteristics such as economic conditions, healthcare infrastructure, and social capital predicted mortality rates, and that some of these contextual associations shifted during the pandemic. If behavior is primarily shaped by such local conditions and norms rather than individual identity, partisans in the same environment should behave similarly; a Democrat in a Republican-majority area should be nearly indistinguishable from her Republican neighbors. Baxter-King et al. (2022) provide the most direct test of this prediction for mask-wearing, finding that Republicans were less likely to wear masks as the Republican share of their zip code increased, while Democratic mask-wearing was unaffected by local partisan context, suggesting that contextual effects on visible health behaviors may operate asymmetrically across partisan groups. This "context dominates" prediction can be tested across multiple dimensions: partisan composition, urbanicity, region, and policy environment. If partisan gaps persist across all of these contextual comparisons, it would provide strong evidence that identity drives behavior independent of local culture, what we term "portable polarization": the idea that partisans carry their behavioral patterns with them regardless of where they live or what surrounds them. Importantly, portable polarization does not require that gaps are identical in every context; it requires only that they persist. Context may shift absolute levels of behavior for all groups without closing the partisan divide, and variation in gap magnitude across contexts is itself analytically informative, potentially revealing which groups are more responsive to local conditions and which are more anchored to nationally

circulated partisan cues.

An alternative within-person explanation emphasizes risk perception: Republicans and Democrats may have arrived at different assessments of pandemic severity, whether through divergent media ecosystems (Baum 2011; Graf and Aday 2008), elite messaging, or identity-driven motivated reasoning, and these different perceptions in turn produced different behaviors. Work in this journal supports the plausibility of this pathway: Hawks (2025) showed that perceived legitimacy of public health authorities and shared health values predicted compliance with COVID-19 directives, suggesting that the perceptual lens through which partisans evaluate health guidance may shape behavioral responses. If this account is correct, partisan gaps should substantially close among partisans who report the same level of concern. However, if stated perceptions are themselves expressions of partisan identity, if partisans report what their group is "supposed" to believe rather than what actually guides their behavior, then perceptions may fail to account for behavioral differences. A related question concerns whether objective pandemic severity can override partisan divergence: if partisan gaps narrow during high-mortality periods and widen when the threat recedes, this would suggest that objective conditions can temporarily override identity-driven behavior.

Taken together, these competing explanations define a logical space within which our empirical analysis operates. The question is not whether partisanship correlates with behavior (prior work establishes that it does) but whether that correlation reflects the independent force of partisan identity or can be reduced to policy, place, demographics, perception, or objective conditions. The identity account generates a distinctive prediction that separates it from all contextual alternatives: if partisan identity shapes health behavior through the mechanisms described above (group conformity, symbolic boundary maintenance, and responsiveness to nationally circulated elite cues) then

its behavioral effects should be *portable*, traveling with individuals regardless of local context, because the sources of partisan influence (national media, party leaders, co-partisan social networks) are not geographically bounded in the way that policy environments, local norms, and demographic compositions are. If gaps persist after each alternative explanation has been given its best chance to account for them, within shared policy regimes, across regions, in both urban and rural settings, and regardless of local partisan composition, the surviving interpretation is that partisan identity has become a fundamental driver of health behavior, one whose influence operates through channels that transcend place.

1.4 The Current Study

Drawing on this competing-explanations framework, we examine two primary research questions. First, are partisan differences in health behaviors (social contacts and mask-wearing, with a descriptive assessment of vaccine acceptance) robust to policy environment, geographic context, and demographic composition? Second, under what conditions are partisan gaps in health behavior largest versus smallest? We test each competing explanation by examining whether partisan gaps narrow where that account predicts they should: within shared policy environments, after demographic adjustment, among partisans with similar risk perceptions, during high-severity periods, and across diverse geographic contexts.

We expect the magnitude of partisan gaps to vary across behaviors and conditions. Behaviors that are visible and low-cost, such as mask-wearing, may be more susceptible to partisan signaling than behaviors that are private and high-stakes, such as reducing social contacts during a mortality surge. Similarly, gaps may narrow when the pandemic's toll is most visible and immediate, as

concrete consequences constrain the space for motivated reasoning. These predictions allow us to test not only *whether* identity matters but *how* it operates differently across behavioral domains and temporal contexts.

We use data from the Berkeley Interpersonal Contact Study (BICS), a nationally representative survey that collected actual counts of daily interpersonal contacts and detailed mask usage during specific interactions across six waves from April 2020 to May 2021. These behavioral measures, concrete counts rather than abstract self-assessments, provide a more direct window into the behaviors most relevant to disease transmission. Our contribution is primarily descriptive: we do not claim to identify causal mechanisms, but we use the pattern of results across multiple contextual comparisons to constrain the set of plausible explanations for partisan behavioral divergence.

2 Data and Methods

2.1 Data

The Berkeley Interpersonal Contact Study (BICS), a nationally representative survey that collected detailed information on social contact patterns, protective behaviors, and demographic characteristics during the COVID-19 pandemic in the United States. BICS recruited respondents from an online panel and used weighting methods to generate nationally representative estimates. Respondents were asked how many people they had face-to-face contact with on the day before the survey interview, and detailed information was collected on up to three of these contacts, including each contact's age, sex, race/ethnicity, mask usage, and physical distance. We analyze waves 1 through 6, spanning April 2020 to May 2021 (N = 18,233). Additional details on survey methodology,

questionnaire design, sampling, and weighting are described in Feehan and Mahmud (2021).

Table 1: Sample Characteristics by Political Party

Characteristic	**Republican**	**Independent**	**Democrat**
N	5,517	4,527	8,189
Mean age (years)	50.0	46.4	46.3
Male (%)	51.5	50.1	46.3
White (%)	89.6	78.4	69.7
College graduate (%)	32.0	29.4	37.1
Metropolitan (%)	82.5	85.4	87.8
Employed (%)	58.6	55.1	59.8
Household size	2.91	2.88	2.87

The three partisan groups differed on several demographic characteristics (Table 1). Republicans were older on average and more likely to be male. The most striking demographic difference was in racial composition: 85% of Republicans identified as white, compared to 71% of Independents and 60% of Democrats. Democrats were more likely to live in metropolitan areas and to hold a college degree. Our analytic sample overrepresents Democrats (44.9%) relative to national estimates of party identification during the study period (30% Democrat, 29% Republican, and 39% Independent; Gallup (2021)) while underrepresenting Independents (24.8% vs. 39%). Republicans are represented at roughly their national share (30.3% vs. 29%).

Survey weights are applied throughout to improve representativeness. These weights adjust for age, gender, race/ethnicity, and education, and we additionally rake on party identification to match national targets. Because partisanship is highly correlated with these demographic characteristics,

the party-adjusted weights produce estimates nearly identical to the original demographic weights for within-party comparisons (see Section 6.1).

Political party identification was assessed by the panel provider prior to the survey. Upon enrollment in the panel, respondents selected their partisan affiliation from standard categories (Democrat, Republican, Independent, or Prefer not to Say/Other). We exclude respondents in the "Prefer not to Say/Other" category to prevent potential misclassification bias. Because party identification was collected independently of our survey instrument and before the pandemic began for most respondents, it is not subject to priming effects from our health behavior questions. Party identification is among the most stable of political attitudes, typically formed in early adulthood and maintained across the life course (Green 2002), making it unlikely that respondents' partisan identification changed meaningfully between the panel provider's measurement and our survey.

Several features of the data warrant consideration. The sample is drawn from an online panel, which may underrepresent populations with limited internet access. Self-reported behavioral measures are subject to social desirability bias and recall error, though the specificity of our measures (actual contact counts for the previous day, mask usage during specific interactions) reduces the scope for such bias relative to abstract attitudinal items. Our assessment of vaccine acceptance is limited to a single wave (May 2021), early in the rollout when eligibility and supply varied across jurisdictions.

2.2 Measures

2.2.1 Dependent Variables

We examine two primary health behaviors measured across all six survey waves, plus vaccination assessed in a single wave.

Social Contacts. Our primary contact measure is the total number of face-to-face contacts respondents reported for the day prior to the survey interview. BICS defined a contact as any in-person interaction involving either physical touch or a conversation of at least a few words. We also examine non-household contacts as a supplementary measure that isolates discretionary social exposure outside the home.

Mask-Wearing. Mask usage is derived from the contact-level alter roster. For each of up to three non-household contacts, respondents indicated whether a face mask was worn during the interaction. We aggregate these reports to construct a respondent-level measure equal to the percentage of non-household contacts during which the respondent wore a mask (0-100 scale). This measure captures reported masking behavior during specific recalled interactions rather than general attitudes toward masks, though it remains subject to recall bias: respondents may be more likely to report contacts that are salient or emotionally close (Marin 2004), and we cannot observe actual masking behavior directly.

Vaccination. Vaccine acceptance, measured in Wave 6 (May 2021) only, is a binary indicator equal to one if the respondent was already vaccinated or stated an intention to vaccinate. Because it was assessed in a single wave, vaccine acceptance is reported descriptively but is not examined in the temporal or contextual analyses; partisan differences are reported in Section 6.4.

2.2.2 Key Independent Variable

Party identification is our primary independent variable. This is a subjective measure: respondents simply indicated which political party they identify with. Respondents were classified as Democrats, Republicans, or Independents based on this self-reported affiliation, collected by the panel provider prior to our survey. Democrats serve as the reference category in all regression models.

2.2.3 Control Variables

Local pandemic severity, measured as lagged county-level COVID-19 incidence and mortality rates, is the most theoretically consequential control, as it captures the objective risk environment to which partisans may differentially respond. We link individual survey responses to geographic data at the ZIP code and congressional district levels to characterize local partisan composition (2020 U.S. House election vote share) and policy environment (county-level mask mandate status, classified as strict, less strict, unspecified, or no mandate). Additional demographic controls include age, gender, race/ethnicity, education, income, and urban/rural residence. These variables are included to test whether partisan gaps reflect demographic sorting rather than the independent effect of identity (see Section 6.2 for detailed results).

2.3 Analytical Strategy

The analysis proceeds in three stages. First, we establish partisan differences in health behaviors and test whether they survive demographic controls. Second, we examine whether risk perception

and objective pandemic severity can account for the gaps. Third, we test whether gaps persist across geographic, policy, and partisan contexts, the central test of "portable polarization."

Our empirical strategy is organized around the competing explanations outlined in the Background. Each explanation predicts that partisan gaps should narrow or disappear under specific conditions; we test each by examining whether gaps persist where that explanation predicts they should not. This pattern-based approach allows us to constrain the set of plausible explanations even though we cannot directly identify causal mechanisms with observational data.

To examine whether partisans differentially respond to changes in pandemic severity, we estimate models interacting party identification with lagged local conditions:

$$Y_{ict} = \beta_1 \text{Party}_i + \beta_2 \log(\text{Incidence}_{c,t-1}) + \beta_3(\text{Party}_i \times \log(\text{Incidence}_{c,t-1})) + \gamma_t + \mathbf{X}_i'\delta + \epsilon_{ict}$$

where Y_{ict} is the outcome for individual i in county c at time t, Party_i indicates partisan affiliation (Democrats as reference), $\log(\text{Incidence}_{c,t-1})$ is the log-transformed county-level incidence rate during the week prior to the survey, γ_t represents wave fixed effects, and $\mathbf{X}_i$ includes individual-level controls. We estimate parallel models substituting mortality for incidence. Both predictors are standardized (z-score) to facilitate comparison. In supplementary analyses, we substitute county fixed effects (γ_c) to isolate within-county temporal variation from all time-invariant county characteristics (see Section 6.3).

For the first research question, we estimate weighted means and regression-adjusted comparisons of health behaviors by party, controlling for demographics. We then stratify by concern level to

test the perception explanation: if partisan gaps disappear among respondents reporting the same level of concern, the perception account is sufficient; if gaps persist, identity may operate through channels beyond stated risk perception.

For the second research question, the wave fixed effects specification above tests whether partisan gaps narrow when objective severity rises, using within-wave cross-county variation in local conditions. Supplementary county fixed effects models absorb all time-invariant county characteristics, isolating within-county temporal variation to ask: as conditions change within the same county, do partisans adjust their behavior differently? We complement this with regional comparisons using estimated marginal means (emmeans) adjusted for survey wave and local mortality, and with stratification by policy environment and local partisan context.

We assess robustness through weight sensitivity analyses comparing results under demographic-only versus party-raked weights (Section 6.1), and supplementary analyses of partisans living in opposing-party congressional districts (Section 6.2.2).

3 Results

We proceed by testing each competing explanation against the empirical patterns, retaining the explanation that survives all tests.

We find substantial partisan differences in social contacts and mask-wearing, as well as in a descriptive assessment of vaccine acceptance (Figure 2). Republicans reported 6.6 total daily contacts compared to 5.3 for Democrats (a gap of 1.2 contacts per day), wore masks during just 54.8% of contacts compared to 65.6% for Democrats (a gap of 11 percentage points), and showed lower vac-

cine acceptance (34.8% vs. 41.1%; see Section 6.4). These differences persisted after controlling for age, gender, race, education, and urban/rural residence, eliminating the possibility that partisan gaps are artifacts of demographic sorting, the first competing explanation. The partisan gap in mask-wearing was larger than differences observed across other demographic categories including race, gender, and education (see Table 4 for behavior levels by demographic subgroup). Detailed contact rates and mask usage levels by partisan group across survey waves, regions, policy environments, and demographic subgroups are reported in Section 6.5. Importantly, these gaps are not driven by respondents with minimal out-of-household activity: restricting the sample to those reporting two or more non-household contacts yields comparable partisan differences in both contacts and mask usage (see Section 6.6.1). Because vaccination was assessed only in Wave 6 and cannot be examined longitudinally, the analyses that follow focus on contacts and mask-wearing.

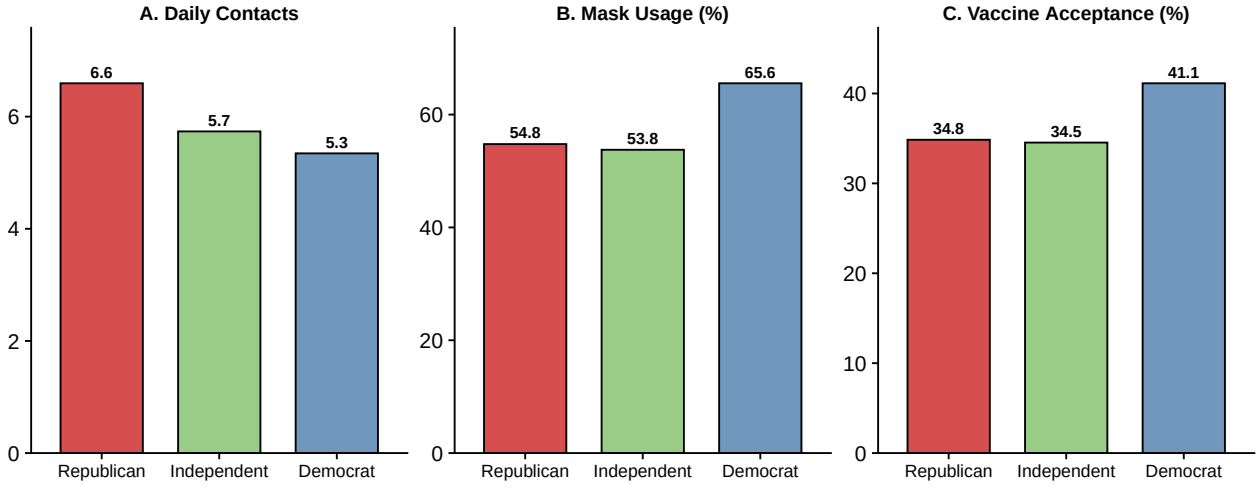

Figure 2: Partisan Differences in Health Behaviors. Panel A shows mean daily contacts, Panel B shows percentage of contacts during which masks were worn, and Panel C shows vaccine acceptance rates (Wave 6 only, May 2021). Vaccine acceptance includes those who were vaccinated or intended to get vaccinated. Values are weighted means using raked survey weights.

Underlying these behavioral differences were substantial partisan gaps in pandemic risk perception (see Section 6.7): 63% of Democrats reported being "very concerned" about COVID-19 compared

to 39% of Republicans, a 23 percentage point gap documented extensively in prior work (Gadarian et al. 2021; Kaushal et al. 2022; Milosh et al. 2021). If these perception differences fully account for the behavioral gaps, then partisans who express the same level of concern should behave similarly. Figure 3 tests this directly by comparing the Republican-Democrat gap within each level of stated concern (see Section 6.7 for levels by party). For social contacts, the partisan gap persists at every level: among the very concerned, Republicans still reported 5.6 daily contacts compared to 5.0 for Democrats. For mask usage, perceived severity does mitigate polarization: the gap narrows as concern increases, and among the "very concerned," Republicans and Democrats report nearly identical mask usage (72% vs. 72%).

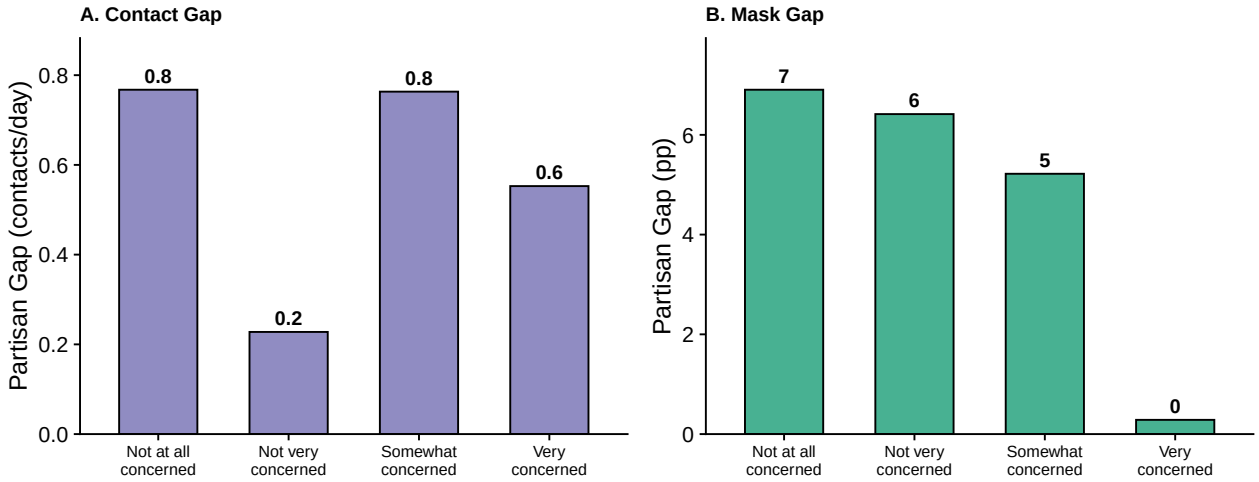

Figure 3: Partisan Gaps in Health Behaviors Within Each Level of Stated Concern. Panel A shows the absolute partisan gap in daily contacts; Panel B shows the absolute partisan gap in mask usage. For contacts, the partisan gap persists at all concern levels. For masks, the gap narrows substantially among the 'very concerned.' Sample sizes: Not at all concerned (n=1,127), Not very concerned (n=1,897), Somewhat concerned (n=5,894), Very concerned (n=9,168).

Interaction models formally confirm these patterns (see Section 6.3). For contacts, the Republican x Very Concerned interaction is statistically significant and negative (β = -1.01, p = 0.03), indicating that concern modestly narrows the partisan contact gap but does not close it. For masks, the interaction is large and significant (β = 12.2, p < 0.001). The mechanism is clear: among the

very concerned, Republicans wore masks during 72% of contacts, nearly identical to Democrats (72%). But this convergence reflects Republicans responding more sharply to concern: Republicans increased mask usage by 43 percentage points from "not at all concerned" to "very concerned," compared to 36 points for Democrats. In sum, perceived severity mitigates partisan polarization for both behaviors, though the effect is substantially larger for mask usage than for social contacts. These results weigh against the perception explanation as a complete account: if partisan behavioral differences were simply downstream of different risk perceptions, they should disappear among partisans who express the same level of concern. They do not, particularly for contacts, where the gap persists at every concern level. Perception matters, but it is insufficient on its own to explain partisan behavioral divergence.

These aggregate gaps were not constant over time. Figure 4 tracks partisan gaps across survey waves alongside local mortality rates. The contact gap was smallest during high-mortality periods, early in the pandemic and during the winter 2020-21 surge (gaps of 0.3-0.7 contacts), and largest when mortality was low (3.5 contacts in September 2020). Regression models confirm that Republicans were more responsive to mortality than Democrats, reducing contacts more sharply when deaths rose (see Section 6.8 for detailed coefficients). The mask gap, by contrast, grew steadily from 6 to 15 percentage points regardless of mortality, showing no convergence during the winter surge. This widening may appear to conflict with the pooled finding that very concerned Republicans and Democrats masked at similar rates, but the two patterns are consistent: the pooled convergence among the very concerned is driven heavily by early waves, and over time the share of Republicans reporting high concern declined steadily, so the population-level mask gap widened even as the within-concern-level gap remained modest (see Section 6.7 for wave-level

breakdowns).

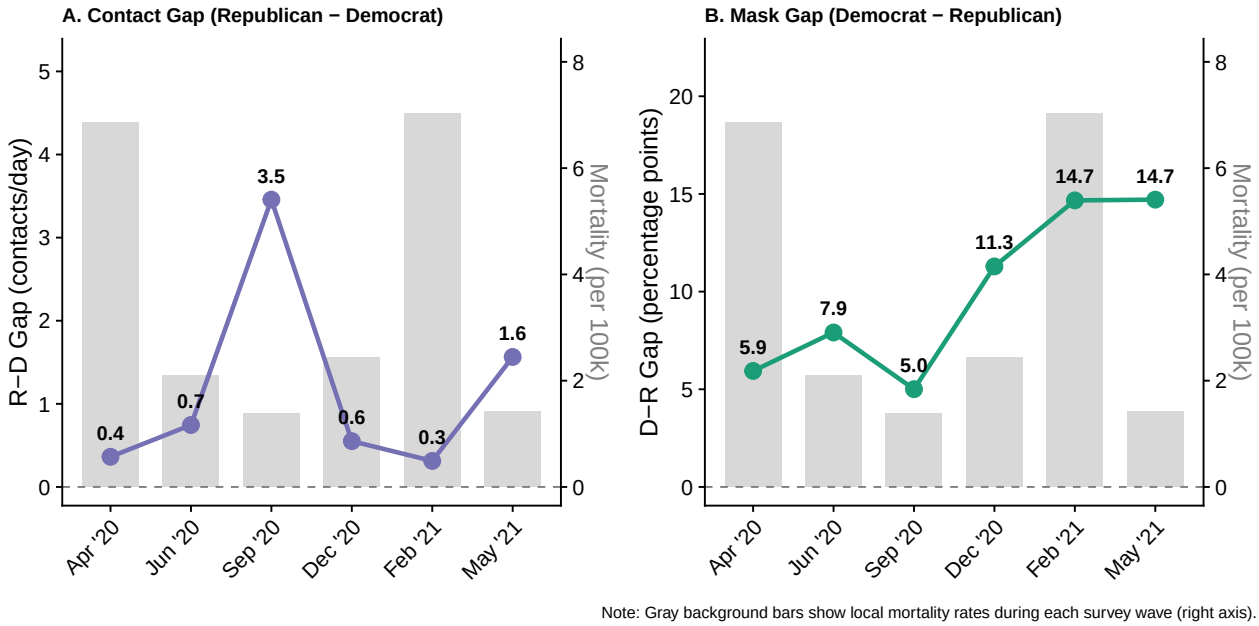

Figure 4: Partisan Gaps in Health Behaviors Across Survey Waves with Mortality Context. Lines show the Republican-Democrat difference in daily contacts (Panel A) and the Democrat-Republican difference in mask usage (Panel B). Gray bars in background show local mortality rates (deaths per 100,000) during each survey wave. The contact gap was smallest when mortality was highest (Feb '21), while the mask gap grew steadily regardless of mortality.

This divergence between contacts and masks offers a partial verdict on the "objective severity overrides identity" explanation. Objective severity can temporarily compress partisan gaps for contacts, a private, high-stakes behavior where the personal consequences of not reducing exposure are immediate, but it does not override identity for mask-wearing, a visible, low-cost behavior that had become an entrenched marker of partisan identity. The most telling evidence comes from the winter 2020-21 surge: Republicans reduced their contacts substantially, converging toward Democratic levels, yet simultaneously reported their *lowest* levels of concern across all waves. Attitudes polarized while behavior converged. This disconnect suggests that expressed attitudes had become performative markers of group allegiance rather than genuine guides to behavior (see Section 6.7.1 for detailed analysis). Neither perception nor severity can fully account for the partisan behavioral

divide.

The final set of tests concerns geographic context. If partisan gaps reflect regional culture or adaptation to local norms, they should narrow or disappear within shared geographic settings. Figure 5 displays Republican-Democrat gaps across census regions, adjusted for survey wave and local mortality. Partisan gaps persist in every region, though with notable variation: the contact gap ranged from 0.9 contacts in the South to 1.7 in the West; the mask gap from 8.3 percentage points in the South to 14.6 in the West. Importantly, where gaps are smaller, as in the South, this reflects Democrats behaving less cautiously (5.5 contacts vs. 4.9 elsewhere), not Republicans becoming more cautious (see Section 6.9 for detailed breakdowns and Section 6.5 for comprehensive behavior levels by party across demographic and contextual subgroups). Context shifts absolute levels of behavior but does not close the partisan gap. City-level comparisons tell the same story: partisan gaps in both contacts and mask usage appear in every BICS sample city (see Section 6.9.6).

These geographic patterns are robust to alternative contextual explanations. Partisan gaps persisted within each mask mandate regime (strict, less strict, and no mandate), eliminating the "it's just policy" explanation (see Section 6.9.1). Gaps remained stable after adjusting for urbanicity and demographics (see Section 6.2). Supplementary analyses of partisans living in opposing-party congressional districts tell a consistent story: while partisans adjust modestly toward local norms, the gap persists in both same-party and cross-party contexts (Section 6.2.2). These findings eliminate the remaining competing explanations: partisan gaps do not disappear among people living under the same mandates, in the same regions, or among members of the opposing party.

Taken together, we can assess the scorecard of competing explanations. Demographic composition does not account for partisan gaps: they persist after controlling for age, gender, race, education,

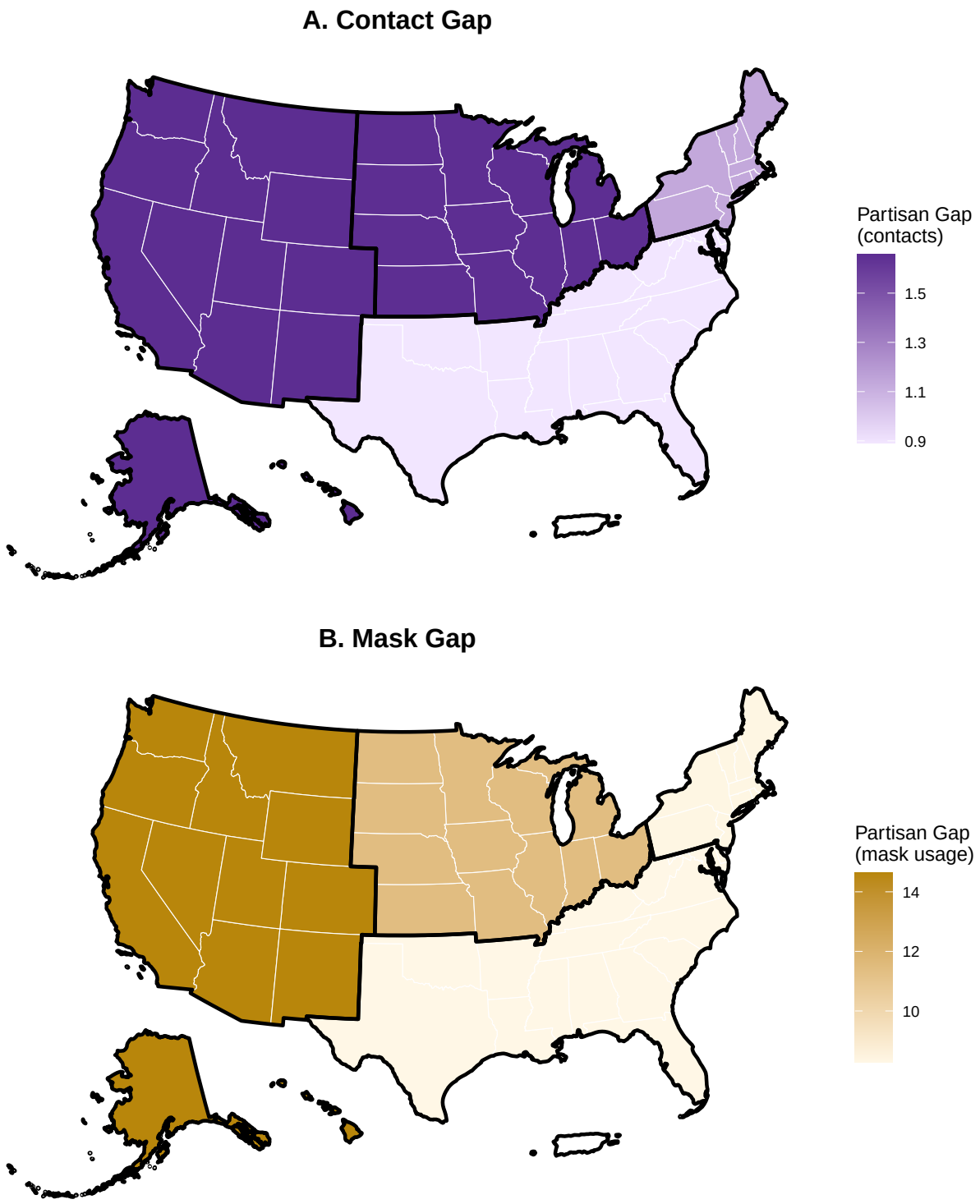

Figure 5: Geographic Distribution of Partisan Gaps by Census Region (Adjusted). Panel A shows the absolute partisan gap in daily contacts; Panel B shows the absolute partisan gap in mask usage. Estimates are adjusted for survey wave and local mortality rates. Darker shading indicates larger gaps. Black borders indicate census region boundaries.

and urbanicity. Information and risk perception offer a partial account for mask-wearing but fail for contacts, where gaps remain at every level of stated concern. Objective severity temporarily compresses contact gaps during mortality surges but leaves mask polarization untouched, and the disconnect between stated concern and behavior during the winter surge undermines the premise that perceptions faithfully guide action. Policy environments do not explain the gaps: they persist within strict, lenient, and no-mandate areas alike. Regional culture shifts absolute levels of behavior but does not close the partisan divide, and even partisans living among members of the opposing party maintain behavioral differences.

The surviving explanation is that partisan identity is itself associated with behavioral polarization, what we term "portable polarization." Partisans carry their behavioral patterns with them across diverse environments, even as they adjust modestly toward local norms. Each competing factor matters at the margins: perception narrows the mask gap among the very concerned, mortality temporarily compresses the contact gap, and regional context shifts absolute levels. But none eliminates the fundamental partisan divide. The gaps are not artifacts of Republicans and Democrats sorting into different environments; they are features of partisan identity itself.

4 Discussion

Partisan gaps in health behavior persisted across every context we examined, a pattern we termed "portable polarization." A long tradition in sociology treats social identities as structuring forces that shape behavior through mechanisms of group conformity, boundary maintenance, and durable dispositions (Bourdieu 1984; Lamont 2000; Tajfel and Turner 1979). In the health domain specif-

ically, Cockerham (2005) argued that health behaviors are not reducible to individual agency but are structured by collective, group-level forces, an insight that health lifestyles research has since confirmed across multiple populations and behavioral domains (Mollborn et al. 2021). Our findings extend this tradition by supporting the claim that partisan identity is a potent structuring force in health behavior. Partisan differences in social contacts and mask-wearing during the COVID-19 pandemic were substantial, with the partisan gap in mask-wearing exceeding differences by race, gender, or education, and proved remarkably resistant to every alternative explanation we tested, consistent with Clinton et al. (2021)'s finding that partisanship substantially outweighed local public health conditions in predicting pandemic mobility behavior. They persisted within shared policy environments, across geographic regions, regardless of local partisan composition, and after controlling for demographics. This pattern of "portable polarization," behavioral differences that prove remarkably robust across diverse contexts even as partisans adjust modestly toward local norms, is consistent with the claim that partisanship has become a fundamental social identity in American life, one that shapes health behavior directly rather than merely correlating with the environments in which partisans are embedded (Hoog and Pat-El 2024).

The specific patterns of variation in partisan gaps offer insight into the mechanisms through which identity may operate. Social contacts, a private, high-stakes behavior, are largely unobservable by one's political community and cannot easily function as signals of partisan allegiance. Contact gaps narrowed during high-mortality periods, suggesting that concrete consequences can temporarily override partisan divergence when the behavior is private and the stakes are immediate. Mask-wearing, a visible, low-cost behavior, showed steadily widening polarization regardless of severity, consistent with its emergence as what Goffman (1963) would recognize as a stigma symbol: a pub-

licly legible marker of group membership, making it a site of impression management rather than a calculated response to risk. Prior work supports this distinction: research on COVID-19 compliance has found that mandatory mask policies shaped not only behavior but social perceptions (Betsch et al. 2020), and identity-protective cognition intensifies rather than diminishes among the most scientifically literate (Kahan 2017). Baxter-King et al. (2022) gesture toward this visibility mechanism, finding that local partisan context shaped Republican but not Democratic mask-wearing. Our findings extend this framework by systematically comparing behaviors that differ in social visibility among the same cross-sectional respondents, a test that prior work was not designed to make. If partisan gaps were driven by generalized health lifestyle dispositions (Cockerham 2005) or divergent risk perceptions, both behaviors should show similar partisan divergence, since both reduce disease transmission. An alternative reading might attribute the divergence to differences in personal cost rather than social visibility, since reducing contacts entails greater material sacrifice than wearing a mask. But this account generates the wrong temporal prediction: if behavioral gaps primarily reflect cost sensitivity, mask gaps should also narrow during mortality surges, since the health cost of not masking rises alongside mortality. Instead, only the private, high-cost behavior responded to heightened risk while the visible, low-cost behavior did not, a pattern more consistent with a social observability mechanism in which the identity costs of deviating from one's group norm offset the health benefits of compliance.

The widening mask gap is consistent with more than one identity-based mechanism: active signaling, in which partisans consciously display or withhold masks to communicate group membership, or the gradual crystallization of group norms, with Democrats habituating to mask-wearing as a default while Republicans consolidated an anti-mask norm. Both are fundamentally identity-driven

but differ in intentionality; the steady, monotonic widening is perhaps more consistent with norm crystallization than with episodic signaling. What our data establish is that whatever the specific mechanism, it operated selectively on the visible behavior and not on the private one, supporting the claim that social observability is a necessary condition for the identity process at work.

The geographic distribution of partisan gaps further illuminates how context and identity interact. Where gaps were smaller, as in the South, the narrowing reflected Democrats behaving less cautiously rather than Republicans becoming more cautious, a pattern visible across demographic and contextual subgroups (Section 6.5): context shifted absolute levels of behavior without closing the partisan divide. Baxter-King et al. (2022) found that local partisan context shaped Republican but not Democratic mask-wearing; our data suggest a more symmetric pattern in which both parties shift toward the behavioral norms of the local political majority, but do so in parallel, leaving the partisan gap intact (see Section 6.2.2). If Republican behavior is primarily shaped by nationally circulated partisan cues and media ecosystems that do not vary much by region, while Democratic behavior responds more to local norms, we would expect exactly the pattern we observe: gaps that persist in every region but vary in magnitude as local conditions pull one group toward the behavioral baseline of its surroundings.

Earlier work using GPS mobility data found that partisan identity moderated responses to elite messaging from the pandemic's earliest days: Democratic-leaning counties responded about twice as strongly to governors' stay-at-home recommendations as Republican-leaning counties, even after accounting for demographics, occupational structure, and local case rates (Grossman et al. 2020). Our individual-level data are consistent with this early divergence but reveal an additional dynamic: the behavioral gaps we measure *widened* over time, particularly for visible behaviors like

mask-wearing, suggesting that what began as differential responsiveness to elite cues crystallized into durable identity markers as the pandemic progressed. The temporal pattern for mask-wearing mirrors what Clinton et al. (2021) documented for stated concern: in their data, Democrats maintained consistently high concern while Republicans became increasingly less concerned over time, with the gap accelerating after partisan elite cues intensified. Yet contact behavior tells a different story: during the winter 2020-21 surge, Republicans reported their lowest concern levels while also reducing contacts substantially, suggesting that expressed attitudes had become performative signals of group allegiance while behavior tracked objective conditions. This finding challenges the assumption, common in survey-based health research, that stated attitudes reliably predict protective behavior on politicized health topics. The persistence of partisan behavioral gaps even after accounting for perceived severity suggests that behavior has decoupled from belief, precisely the space where identity-based theories do their explanatory work (Goffman 1963; Lamont 2000; Tajfel and Turner 1979). Experimental evidence reinforces this point: Pink et al. (2021) found that Republican elite endorsements increased vaccination intentions among Republicans, while Democratic endorsements produced backlash, reducing Republicans' willingness to encourage others to vaccinate, suggesting that partisan identity filters even well-intentioned health communication.

This study contributes to the sociology of health by showing that partisan identity is independently associated with health behavior in ways not reducible to policy, perception, or place, extending recent work in this journal showing that partisan polarization structures trust in scientific authorities in ways that intensify among the most knowledgeable (Gauchat 2025) from the domain of attitudes to the domain of behavior. The competing-explanations framework offers a generalizable approach for descriptive studies seeking to constrain causal accounts: by systematically testing where each

alternative explanation predicts gaps should narrow, we identify the account that survives the most stringent comparisons.

Several limitations warrant consideration alongside these contributions. Our data are observational and cross-sectional within each wave, precluding strong causal claims; we observe associations between partisan identity and health behavior but cannot definitively rule out unmeasured confounders. All behavioral measures are self-reported, and Yair and Huber (2020) have shown that survey items can confound sincere partisan differences with expressive responding. If such dynamics operate in our data, we would be capturing identity signaling in the survey response itself, though notably, this would constitute the same identity-driven mechanism we theorize about, operating at the level of response rather than behavior. Additionally, Clinton, Lapinski, and Trussler (2022) documented differential partisan nonresponse in surveys conducted during this period, with Republicans, particularly those least engaged with public health guidance, being less likely to participate. If this selection operates in our data, the Republicans in our sample are likely more moderate than the Republican population as a whole, meaning our estimated partisan gaps represent a conservative lower bound on the true behavioral divergence (see Section 6.1 for further discussion). The specificity of our measures (contact counts for the previous day, mask usage during specific recalled interactions) reduces the scope for expressive responding, and the contact measure provides a partial check: counting yesterday's face-to-face conversations is a concrete recall task with less room for motivated reasoning than reporting whether one "usually" wears a mask, and partisan gaps in contacts persist even during high-mortality periods when expressive incentives to minimize concern were strongest among Republicans. Our survey waves span April 2020 to May 2021, capturing a critical period but not the later Delta and Omicron waves or the longer arc

of vaccine polarization; Wallace et al. (2023) documented that the partisan mortality gap widened substantially after vaccines became available, suggesting that the behavioral divergence we observe likely intensified beyond our observation window. These temporal boundaries are also a strength: by examining a period when partisan scripts were actively forming, we observe polarization as it developed rather than after it crystallized.

Each limitation suggests a direction for future work. Longitudinal designs following the same individuals over time could establish whether partisan identity causes behavioral change or whether partisan self-sorting into different behavioral patterns precedes identity consolidation. Comparative studies across health crises, including non-pandemic contexts such as gun violence, environmental health, or reproductive health, could test whether portable polarization is specific to COVID-19 or characterizes partisan identity's influence on health behavior more broadly; evidence that ideology shaped vaccine hesitancy even before the pandemic (Baumgaertner, Carlisle, and Justwan 2018) suggests that the dynamics we document may have deeper roots. Experimental work could examine whether making mortality consequences salient narrows partisan gaps in real time, providing a causal test of the severity mechanism we observe descriptively. Finally, network-based studies could illuminate the social pathways through which partisan behavioral norms are transmitted and reinforced. Mosleh et al. (2021) found that shared partisanship dramatically increases social tie formation, suggesting that partisan homophily may create echo chambers for health behavior norms that amplify the divergence we document.

The same dynamics of group conformity, symbolic boundary-making, and habitus that sociologists have long studied in the contexts of race, religion, and class now operate through partisanship in the health domain. Understanding how to bridge these divides during health crises requires not

just better information or clearer messaging, but a sociological understanding of why people adopt the health behaviors their group expects of them, even when those behaviors may not serve their material interests.

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6 Appendix

6.1 APPENDIX A. Weight Sensitivity Analysis

To assess the robustness of our findings to different weighting approaches, we compare results across three scenarios: (1) unweighted estimates, (2) original demographic weights that adjust for age, gender, race/ethnicity, and education, and (3) raked weights that additionally adjust for party identification to match national targets from Gallup polling (30% Democrat, 29% Republican, 39% Independent; Gallup (2021)). The raking procedure adjusts for potential correlations between party identification and demographic characteristics in our sample.

A key finding from this analysis is that raked weights and original demographic weights produce identical within-party estimates for all outcomes. This equivalence occurs because the raking procedure applies a constant multiplier to all observations within each party group. When calculating within-party means or party-specific regression coefficients, this constant multiplier cancels out. Raking only matters for population-level estimates that pool across partisan groups, where it adjusts the relative contribution of each party to the overall estimate.

We observe small differences between unweighted and weighted estimates. The Republican-Democrat contact gap is 1.33 contacts per day using unweighted data compared to 1.25 contacts per day with demographic weights—a 6% reduction. The vaccine acceptance gap is -18.6 percentage points (unweighted) versus -17.0 percentage points (weighted)—a 9% reduction in the absolute gap. Mask usage differences remain stable across all weighting scenarios, with the Republican-Democrat gap at approximately -10.8 percentage points regardless of weights used.

Importantly, all partisan differences remain highly significant ($p < 0.001$) across all three weighting

scenarios. These results demonstrate that our findings are robust to the choice of weighting specification and that the substantive conclusions about partisan gaps in health behaviors do not depend on how we adjust for sample composition.

6.1.1 Direction of Potential Selection Bias

Beyond weighting, a distinct concern is that participation in an online panel survey may itself be correlated with partisanship in ways that bias our estimates. Clinton et al. (2022) documented that Republicans were less likely than Democrats to participate in surveys conducted during the pandemic, and that this differential nonresponse was concentrated among the least politically engaged and least health-conscious Republicans — precisely those most likely to have maintained high contact rates and low mask usage. If this pattern extends to the BICS sample, the Republicans in our data would be more compliant and health-conscious than the Republican population as a whole, meaning that the partisan gaps we estimate are likely *attenuated* relative to the true population differences.

The logic runs as follows. Willingness to participate in a university-administered health survey plausibly selects for civic engagement, health consciousness, and openness to public health messaging. Among Democrats, these traits are broadly aligned with partisan norms, so selection likely has modest effects on the composition of Democratic respondents. Among Republicans, however, these traits are in tension with partisan signals that framed pandemic precautions as overreach; the Republicans who opted into the survey are therefore plausibly those most willing to deviate from the strongest partisan behavioral scripts. The most disengaged Republicans — those who dismissed COVID-19 precautions most thoroughly — are the least likely to appear in our sample. This asym-

metric selection would compress the observed partisan gap from the Republican side, making our estimates conservative.

This reasoning is consistent with the weight sensitivity results above: adjusting for observable demographic imbalances produces only modest changes in partisan gaps, suggesting that the respondents who *did* participate are reasonably representative on observable characteristics. But to the extent that unobservable characteristics (e.g., distrust of institutions, disengagement from health monitoring) are correlated with both nonparticipation and extreme behavior, the true gaps are likely larger than what we report.

6.1.2 Summary Table

Table 2: Partisan Differences in Health Be

Outcome	Comparison	Unweighted	Original Weights
Contacts	Republican - Democrat	1.33 (0.17)***	1.25 (0.17)***
Contacts	Independent - Democrat	0.28 (0.18)NA	0.39 (0.17)*
Mask Usage	Republican - Democrat	-10.79 (0.90)***	-10.78 (0.92)***
Mask Usage	Independent - Democrat	-11.72 (0.99)***	-11.79 (1.00)***
NA	Republican - Democrat	-0.19 (0.02)***	-0.17 (0.02)***
NA	Independent - Democrat	-0.14 (0.02)***	-0.13 (0.02)***

Note: Estimates shown with standard errors in parentheses. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.00$

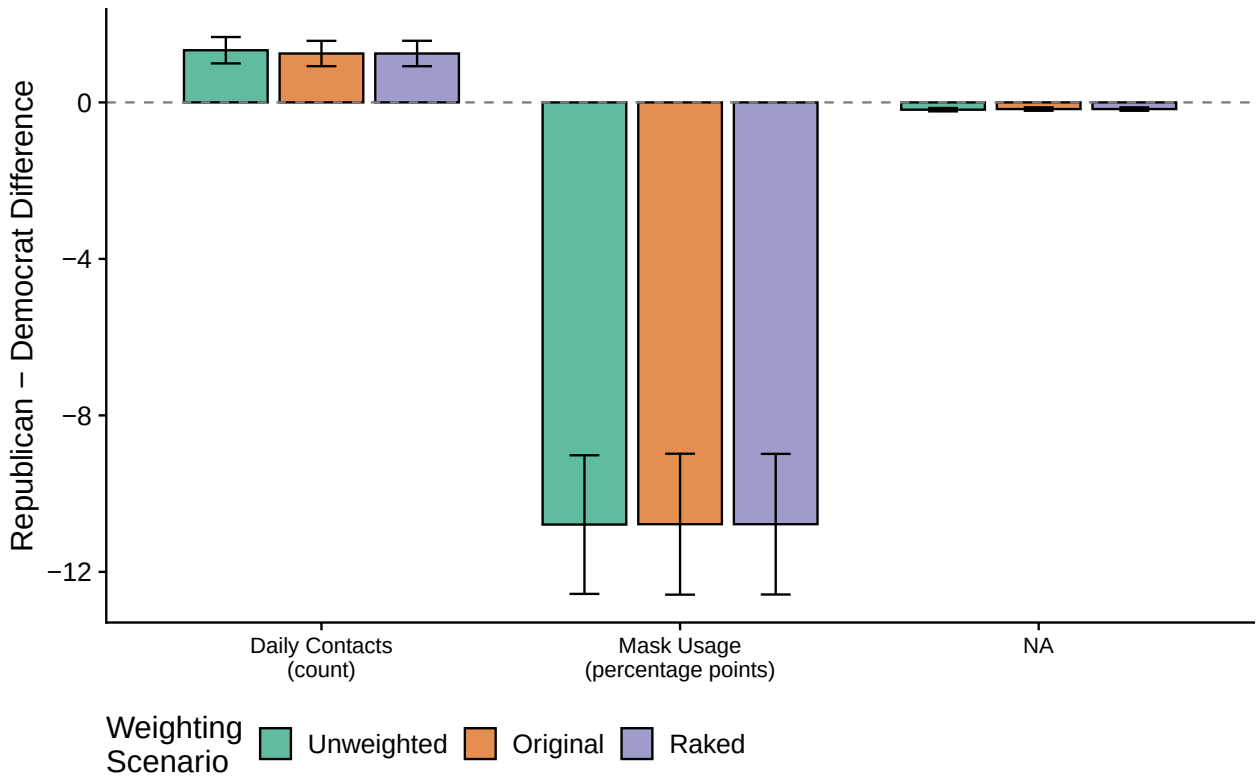

Figure 6: Republican-Democrat Differences in Health Behaviors Across Weighting Scenarios

6.1.3 Comparison of Partisan Gaps

6.2 APPENDIX B. Robustness to Context and Demographic Controls

This section presents a formal test of whether partisan gaps in health behaviors can be explained by local context (metropolitan status, county mask mandates) or demographic composition (sex, race, age, education). We estimate partisan gaps by census region using three nested models and examine whether gaps shrink toward zero as controls are added.

The results show that partisan gaps in both contacts and mask usage remain substantial even after controlling for local context and demographics. For contacts, the gaps are remarkably stable across specifications, suggesting that neither urban-rural differences, policy environments, nor demographic composition can explain partisan behavioral differences. For masks, gaps shrink

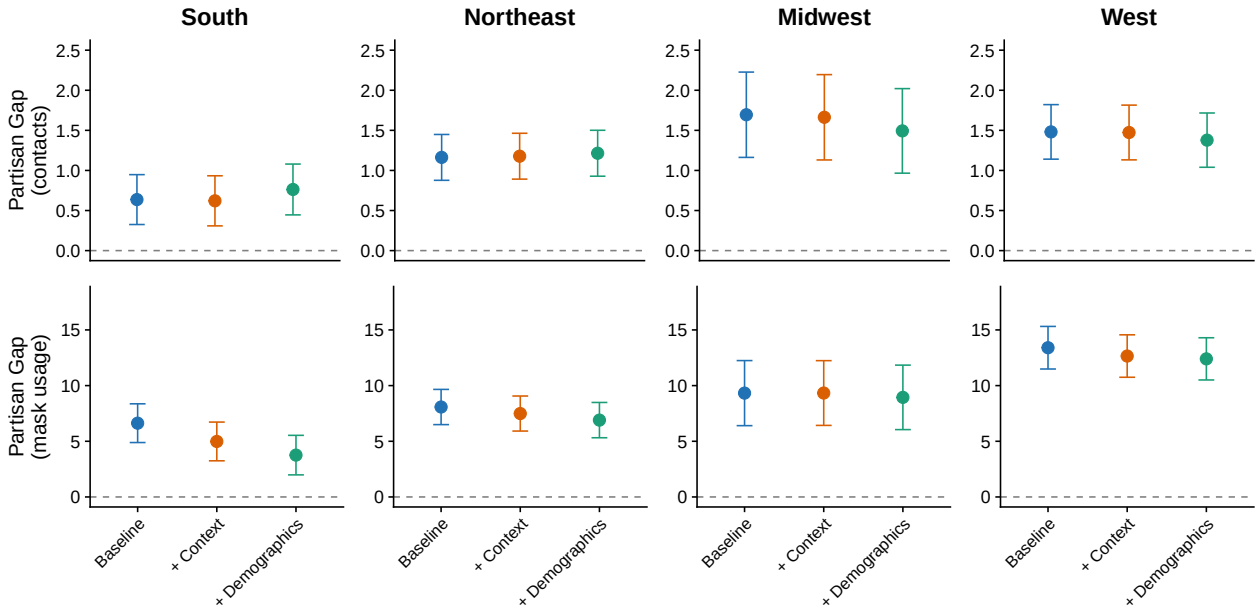

Figure 7: Effect of Controlling for Local Context and Demographics on Regional Partisan Gaps. Top row shows the absolute partisan gap in daily contacts (N = 12,753); bottom row shows the absolute partisan gap in mask usage (N = 9,087). Sample restricted to Democrats and Republicans; mask usage sample is smaller because it is conditional on reporting at least one contact with mask information. Each panel represents a census region. Points show gaps from three nested models: Baseline (wave + mortality), + Context (adds metro status and county mask mandates), and + Demographics (adds sex, race, age, and education). If context or demographics explain partisan gaps, the gaps should shrink toward zero as controls are added.

modestly—particularly in the South—but persist in all regions even in the fully adjusted model, with gaps ranging from approximately 4 percentage points (South) to 12 percentage points (West).

6.2.1 Partisan Gaps by Congressional District Context

Figure 8 presents a complementary view of the main text analysis (Figure 9), showing the absolute Republican-Democrat gaps in contacts and mask usage by congressional district partisan context across the three model specifications.

The gaps remain stable across model specifications in both district contexts. In Democratic-led districts, the contact gap ranges from 1.4-1.6 contacts across specifications; in Republican-led districts, it ranges from 0.5-0.7 contacts. For masks, gaps in Democratic districts range from 7-8 percentage points, while gaps in Republican districts range from 6-8 percentage points. These results complement the main text finding that partisan gaps persist regardless of local partisan context.

6.2.2 Robustness to Model Specification

Figure 9 displays predicted health behaviors by party and congressional district context, showing that partisan gaps persist in both Democratic-led and Republican-led districts.

Figure 10 demonstrates that partisan gaps in health behaviors persist regardless of whether we control for local context and demographic characteristics. The figure compares unadjusted estimates (wave only) with adjusted estimates (controlling for incidence, mask mandates, metro status, and demographics).

The adjustment for context and demographics has minimal impact on partisan gaps. For contacts, the Republican-Democrat gap shifts only slightly when adding controls: in Democratic districts,

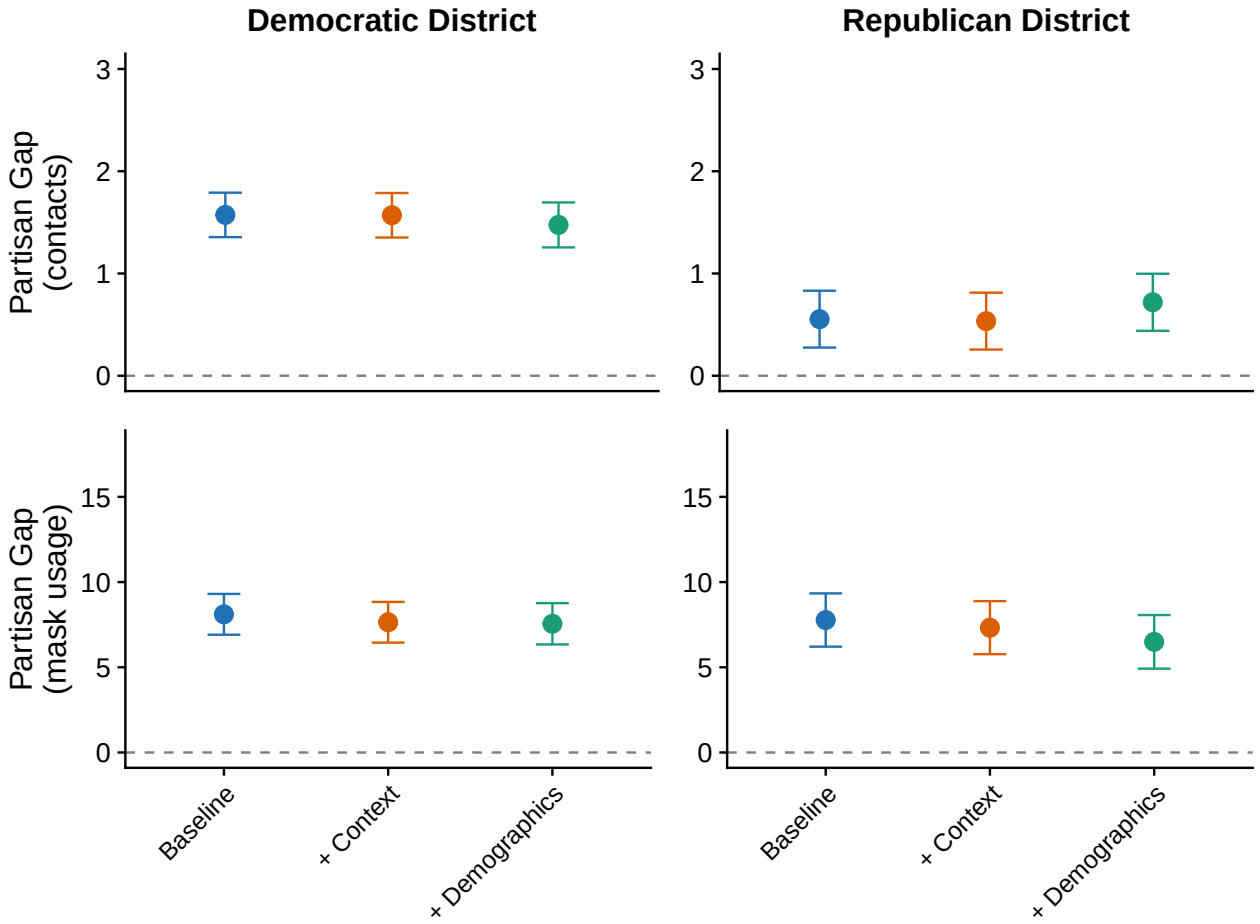

Figure 8: Partisan Gaps by Local Partisan Context Across Model Specifications. Left column shows gaps in Democratic-led congressional districts; right column shows gaps in Republican-led districts. Top row shows the absolute partisan gap in daily contacts; bottom row shows the absolute partisan gap in mask usage. Points show gaps from three nested models: Baseline (wave + mortality), + Context (adds metro status and county mask mandates), and + Demographics (adds sex, race, age, and education). Sample restricted to Democrats and Republicans.

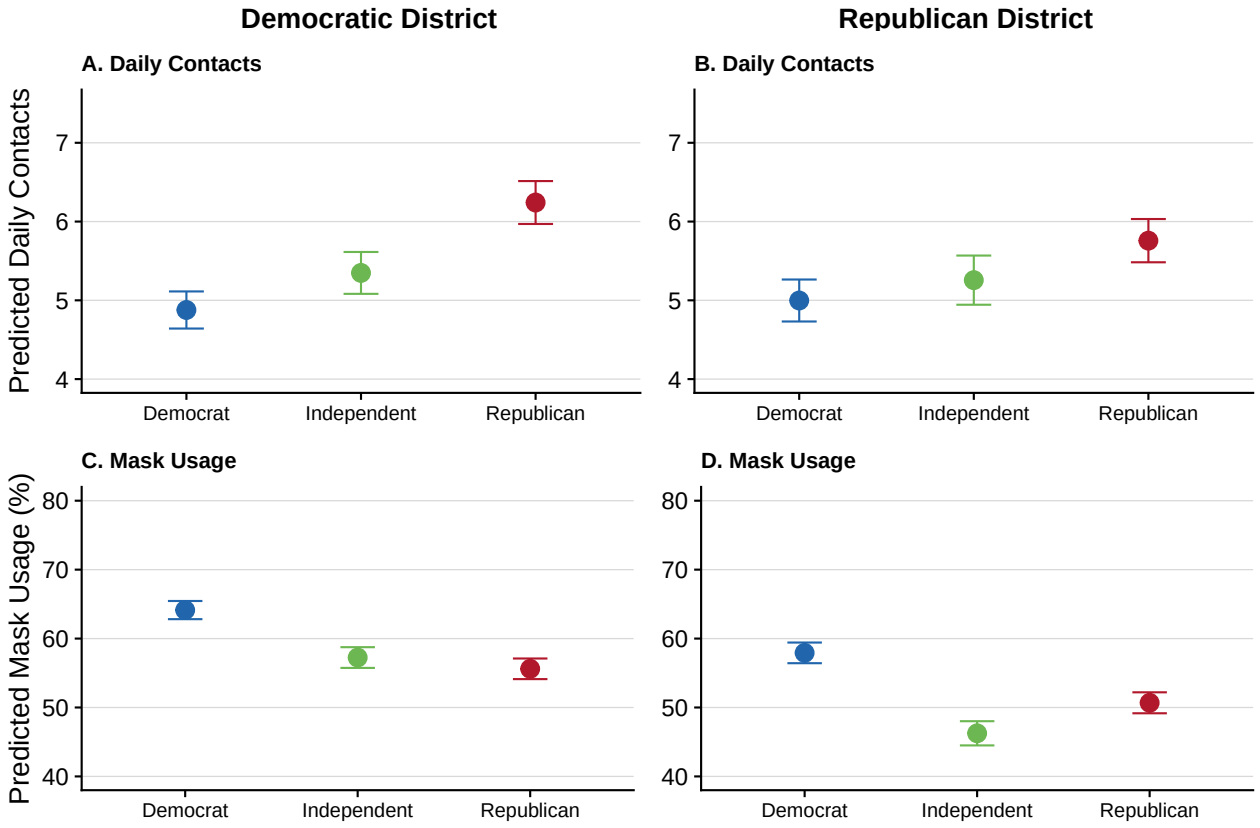

Figure 9: Predicted Health Behaviors by Party and Congressional District Context. Panels A-B show predicted daily contacts; Panels C-D show predicted mask usage. Left column shows Democratic-led districts; right column shows Republican-led districts. Estimates are adjusted for local incidence, mask mandates, metro status, and demographics. Partisan gaps persist in both district contexts.

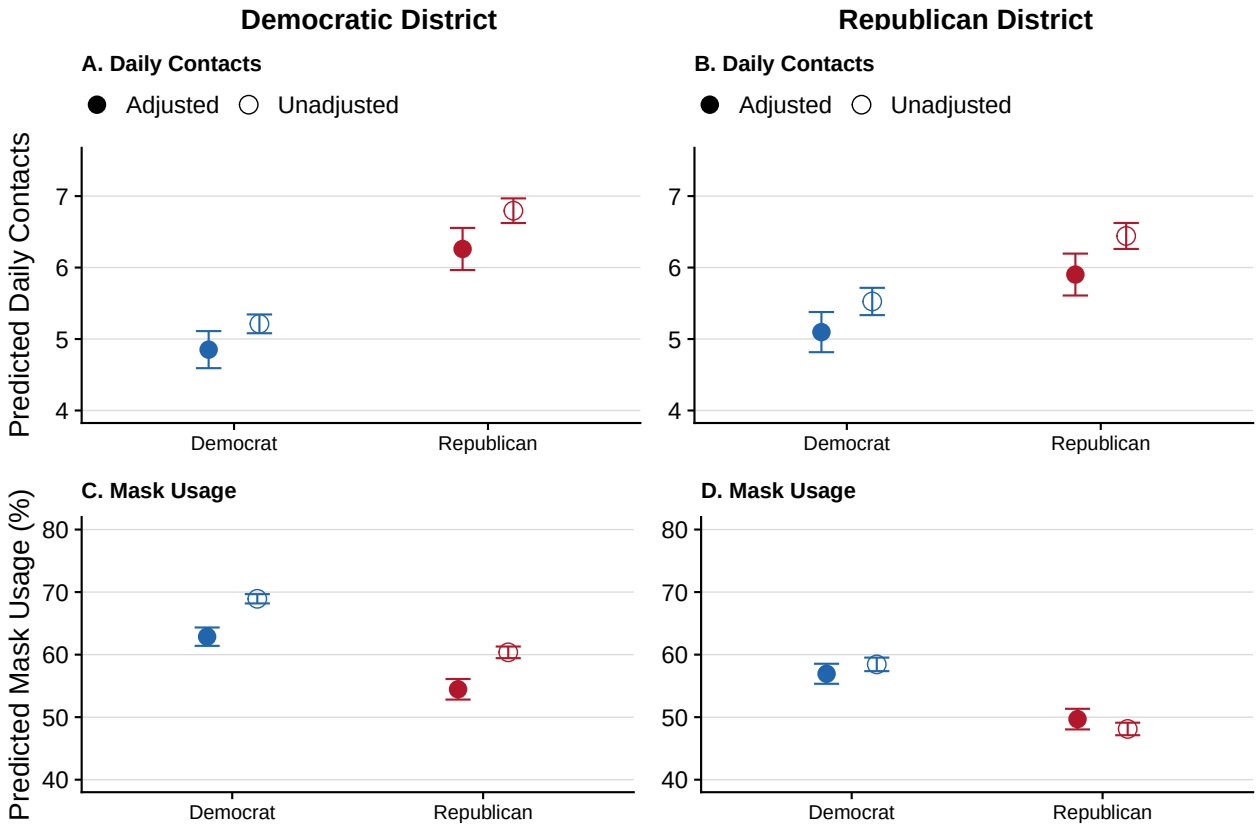

Figure 10: Robustness of Partisan Gaps to Model Specification. Panels A-B show predicted daily contacts; Panels C-D show predicted mask usage. Left column shows Democratic-led districts; right column shows Republican-led districts. Open circles show unadjusted estimates (wave only); filled circles show adjusted estimates (controlling for incidence, mask mandates, metro status, demographics). Partisan gaps persist regardless of model specification.

the gap remains approximately 1.3-1.5 contacts whether unadjusted or adjusted; in Republican districts, the gap remains approximately 0.5-0.8 contacts. For masks, the Democratic advantage over Republicans remains stable at approximately 7-10 percentage points across both district contexts and model specifications. This stability indicates that partisan differences in health behaviors are not explained by compositional differences or local contextual factors.

6.3 APPENDIX C. Interaction Models: Concern and Mortality as Moderators

To formally test whether stated concern and objective mortality rates moderate partisan gaps in health behaviors, we estimate interaction models of the form:

$$\text{Behavior} = \beta_0 + \beta_1(\text{Republican}) + \beta_2(\text{Concern}) + \beta_3(\text{Mortality}) + \beta_4(\text{Rep} \times \text{Concern}) + \beta_5(\text{Rep} \times \text{Mortality}) + \text{FE}$$

In this specification, the partisan gap equals $\beta_1 + \beta_4(\text{Concern}) + \beta_5(\text{Mortality})$. A negative β_4 indicates that higher concern *narrows* the partisan gap; a negative β_5 indicates that higher mortality narrows it. Concern is measured as a binary indicator for reporting being "very concerned" about COVID-19; mortality is the standardized log of the previous week's county-level mortality rate.

We estimate two versions of this model to distinguish between national/temporal patterns and local variation:

1. **Wave FE model**: Includes wave fixed effects and controls for metropolitan status. This

captures how the partisan gap varies with *national* pandemic severity (high-mortality waves vs. low-mortality waves).

2. **County FE model**: Includes county fixed effects. This asks: *within the same county*, when local mortality rises, does the partisan gap narrow? This isolates the effect of local conditions from all time-invariant county characteristics.

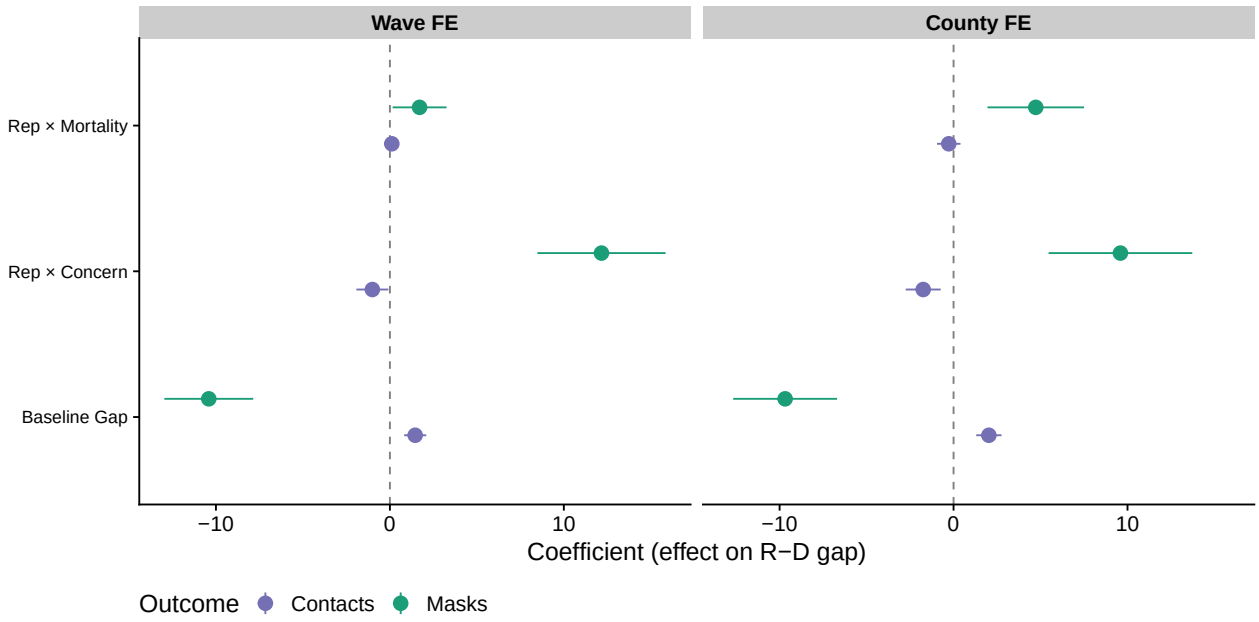

Figure 11: Moderators of Partisan Gaps: Wave FE vs. County FE Models. Points show regression coefficients from interaction models predicting contacts (purple) and mask usage (green). Left panel uses wave fixed effects (national/temporal variation); right panel uses county fixed effects (local variation). 'Baseline Gap' is the Republican main effect. 'Rep × Concern' shows how being very concerned changes the gap. 'Rep × Mortality' shows how a 1 SD increase in local log mortality changes the gap. Negative values indicate the gap narrows. Error bars are 95% CIs.

Figure 11 displays the results from both specifications.

Wave FE Model (National/Temporal Patterns): For contacts, the baseline Republican-Democrat gap is 1.45 contacts per day. The concern interaction is statistically significant and negative (β = -1.01, p = 0.03), indicating that stated concern does modestly narrow the partisan contact gap. The

mortality interaction is negative (β = 0.11, p = 0.573): during high-mortality waves, the partisan gap narrows.

For masks, the baseline gap is 10.4 percentage points. The concern interaction is large and significant (β = 12.2, p < 0.001): being "very concerned" substantially narrows the mask gap. The mortality interaction is smaller and in the opposite direction (β = 1.7).

County FE Model (Local Variation): The county FE model asks whether the partisan gap narrows *within counties* when local mortality rises. For contacts, the mortality interaction is -0.28 (p = 0.424), indicating that local mortality conditions also moderate the partisan gap. For masks, the local mortality interaction is 4.7 (p = 0.001).

Both specifications support the same conclusion: **concern moderates the mask gap but not the contact gap; mortality moderates the contact gap but not the mask gap.** This pattern holds whether we examine national trends (wave FE) or local variation (county FE), suggesting it reflects a genuine behavioral response to pandemic conditions rather than a compositional artifact.

6.4 APPENDIX D. Vaccine Acceptance

Vaccine acceptance, measured during the final survey wave in May 2021, showed pronounced partisan differences. We define vaccine acceptance as being vaccinated or intending to get vaccinated, which captures underlying attitudes toward vaccination independent of differential access during the early rollout period. Among Democrats, 41.1% accepted vaccines, compared to 34.5% of Independents and 34.8% of Republicans—a 6 percentage point gap. Because vaccination was assessed in a single wave, it cannot be examined temporally or geographically with the same depth as contacts and mask-wearing; this finding should be interpreted as a cross-sectional snapshot from

May 2021, early in the U.S. rollout when supply constraints and eligibility tiers varied substantially across states.

6.5 APPENDIX E. Contact Rates and Mask Usage by Partisan Group: Detailed Behavior Levels

To complement the gap-focused analyses in the main text, this section presents weighted mean behavior levels by party across all contextual breakdowns used in the paper, including survey wave, census region, policy environment, and demographic subgroup. These tables may be of particular interest to researchers modeling disease transmission in politically heterogeneous populations, as they provide group-specific contact rates and mask usage estimates that can parameterize differential mixing and protective behavior in epidemiological models.

Table 3: Weighted Mean Behavior Levels by Party and Context

	Contacts			Mask Usage (%)			
Context	Rep.	Ind.	Dem.	Rep.	Ind.	Dem.	N
Overall							
Pooled	6.63	5.74	5.34	55.2	53.9	65.7	18,233
Survey Wave							
Apr ’20	4.10	4.34	3.74	54.2	44.2	60.2	2,355
Jun ’20	5.67	4.93	4.92	51.8	44.5	59.7	2,180
Sep ’20	8.64	6.04	5.19	64.7	66.6	69.7	3,040
Dec ’20	5.78	5.47	5.23	61.4	61.5	72.7	2,837
Feb ’21	5.95	4.84	5.64	57.6	64.5	72.3	2,827
May ’21	7.71	7.23	6.14	45.1	46.7	59.9	4,994
Census Region							
Northeast	6.42	5.69	5.32	60.7	61.5	69.3	6,341
Midwest	6.98	6.11	5.28	48.1	42.1	59.8	1,749
South	6.66	6.03	5.80	52.8	44.9	61.1	4,979
West	6.55	5.32	4.93	52.6	57.1	67.2	5,063
Metro Status							
Metropolitan	6.49	5.46	5.24	57.4	56.3	66.9	15,796
Non-Metropolitan	6.54	7.36	6.24	47.8	42.1	54.7	1,395
Mask Mandate							
No Mandate	7.02	6.67	5.57	45.4	42.3	54.0	2,328
Less Strict	6.84	4.81	5.81	60.4	63.0	71.3	3,641
Strict	6.39	5.71	5.13	56.1	55.1	66.3	10,433
Sample City							
Atlanta	7.26	6.18	5.36	50.1	42.5	56.9	1,673
Bay Area	6.64	5.81	4.89	58.7	58.0	71.8	1,654
Boston	6.21	6.30	4.89	56.5	59.2	67.8	1,655

NY	6.00	4.82	5.23	66.3	65.5	74.0	1,698
National	6.58	5.66	5.63	53.5	49.8	64.2	8,256
Philadelphia	6.44	5.98	4.94	56.8	62.1	65.8	1,700
Phoenix	6.68	5.09	5.40	51.8	55.6	61.2	1,597
Opposing-Party CD							
Opposing-party CD	6.74	5.74	5.34	60.5	53.7	65.5	15,844
Same-party CD	6.44	4.60	5.28	48.1	67.5	82.5	2,389

Across both tables, Democrats consistently report fewer daily contacts and higher mask usage than Republicans in every contextual breakdown. The partisan ordering (Democrat < Independent < Republican for contacts; Democrat > Independent > Republican for masks) holds with notable consistency across survey waves, census regions, metropolitan and non-metropolitan areas, mask mandate environments, sample cities, and same- vs. opposing-party congressional districts. The absolute levels also reveal meaningful variation across contexts—for instance, contacts are higher in later waves as the pandemic progressed, and mask usage is lower in areas without mask mandates—but the partisan ranking remains stable throughout.

Table 4 presents the same weighted mean behavior levels stratified by the demographic and socioeconomic covariates included in our regression models. This complements the contextual breakdown above by showing that partisan differences persist across age groups, racial and ethnic categories, gender, education levels, employment status, household sizes, and income brackets.

Table 4: Weighted Mean Behavior Levels by Party and Demographic/SES Category

	Contacts			Mask Usage (%)			
Category	Rep.	Ind.	Dem.	Rep.	Ind.	Dem.	N
Age Group							
[18,30)	8.13	6.36	6.19	51.4	51.1	61.1	3,689
[30,45)	8.05	6.92	6.33	61.6	54.2	68.3	6,146
[45,55)	7.35	5.61	6.77	57.8	56.7	71.6	2,998
[55,65)	6.03	5.81	4.03	49.2	53.4	62.6	2,570
[65,100]	4.03	3.64	3.01	50.8	54.0	62.9	2,830
Race							
Asian	4.59	4.21	5.15	68.9	65.6	69.4	1,208
Black	6.41	5.58	4.88	61.3	63.2	67.1	1,976
Other / Mixed	5.38	5.34	5.24	52.7	49.2	66.4	736
White	6.72	5.89	5.49	54.2	52.4	64.9	14,313
Hispanic Ethnicity							
Hispanic	6.60	6.44	5.77	56.6	51.0	65.1	1,960
Not Hispanic	6.59	5.62	5.24	54.5	54.2	65.7	16,273
Gender							
Female	6.07	5.65	5.22	50.9	49.4	62.8	9,358
Male	7.09	5.82	5.49	58.0	57.9	68.5	8,875
Education							
College graduate	6.97	5.82	5.90	63.1	58.4	71.1	9,272
High school graduate	6.02	5.74	4.85	51.5	48.8	60.3	3,101

Non-high school graduate	7.43	4.52	5.25	40.6	51.9	59.5	386
Some college	6.82	5.84	5.21	50.1	54.9	64.1	5,474
Employment Status							
Not Working	5.16	3.65	3.58	50.3	52.3	61.6	5,541
Working	8.01	7.64	6.70	57.0	56.5	68.0	11,457
Household Size							
1 (Lives alone)	4.18	3.87	3.62	52.5	50.7	65.2	3,293
2	5.07	4.28	3.57	49.5	53.5	60.3	5,603
3	7.03	5.94	5.40	56.4	53.7	69.4	3,735
4+	9.10	8.25	8.12	60.4	55.7	68.4	5,602
Household Income							
Less than $30,000	5.82	5.14	4.76	48.9	53.2	61.8	2,444
$30,000 to $59,999	6.29	6.14	4.65	49.2	52.0	65.2	4,190
$60,000 to $89,999	6.99	5.72	5.15	53.9	58.1	65.5	2,933
$90,000 to $199,999	7.12	6.72	6.24	61.3	56.4	71.3	4,757

The partisan ordering—Republicans reporting more contacts and lower mask usage than Democrats—holds across every demographic subgroup. Notably, the gap persists among both college graduates and non-college respondents, among both working and non-working individuals, and across all income brackets. Absolute levels vary meaningfully by demographics: younger respondents and those in larger households report more contacts, while college graduates and higher-income respondents report higher mask usage. Yet within every subgroup, the Republican-Democrat difference remains clearly visible.

6.6 APPENDIX F. Non-Household Contacts and Distributional Robustness

A potential concern with the total-contacts measure is that partisan differences could reflect household composition rather than discretionary social behavior—for example, if Republicans live in larger households. A related concern is that the mean contact count could be pulled by outliers in the right tail, inflating the apparent partisan gap. This appendix addresses both issues by (1) decomposing contacts into household and non-household components and (2) estimating the partisan gap using negative binomial models and quantile regressions that are robust to distributional assumptions.

Figure 12 shows the decomposition of mean daily contacts into household and non-household components by party, pooled across all survey waves. The Republican–Democrat gap in total contacts (1.29 contacts/day) is overwhelmingly driven by non-household contacts (1.21 contacts/day, or 94% of the total gap), with household contacts contributing only 0.07 contacts/day. This indicates that the partisan contact gap reflects differences in discretionary social behavior—not household composition.

Figure 13 shows non-household contacts by party across survey waves. The partisan gap in discretionary contacts is present at every wave, generally widening during fall 2020 (wave 3) before narrowing again.

Finally, Figure 14 presents the estimated partisan gap from multiple model specifications, demonstrating that the Republican–Democrat difference is robust to distributional assumptions. The negative binomial models account for overdispersion common in count data, while quantile regressions show that the gap is not driven by extreme values in the right tail—it is present at the 25th, 50th, and 75th percentiles of the contact distribution.

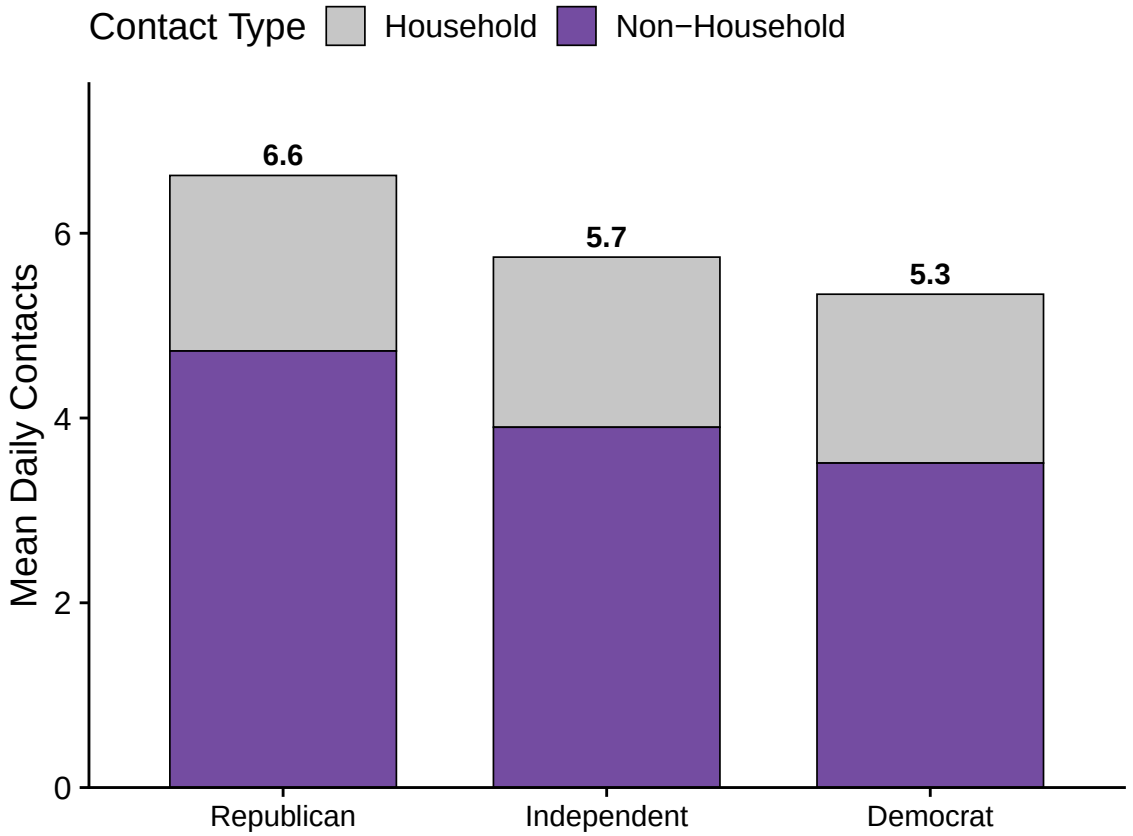

Figure 12: Decomposition of Daily Contacts into Household and Non-Household Components by Party. Bars show weighted mean contacts pooled across all survey waves. The partisan gap is driven almost entirely by non-household (discretionary) contacts.

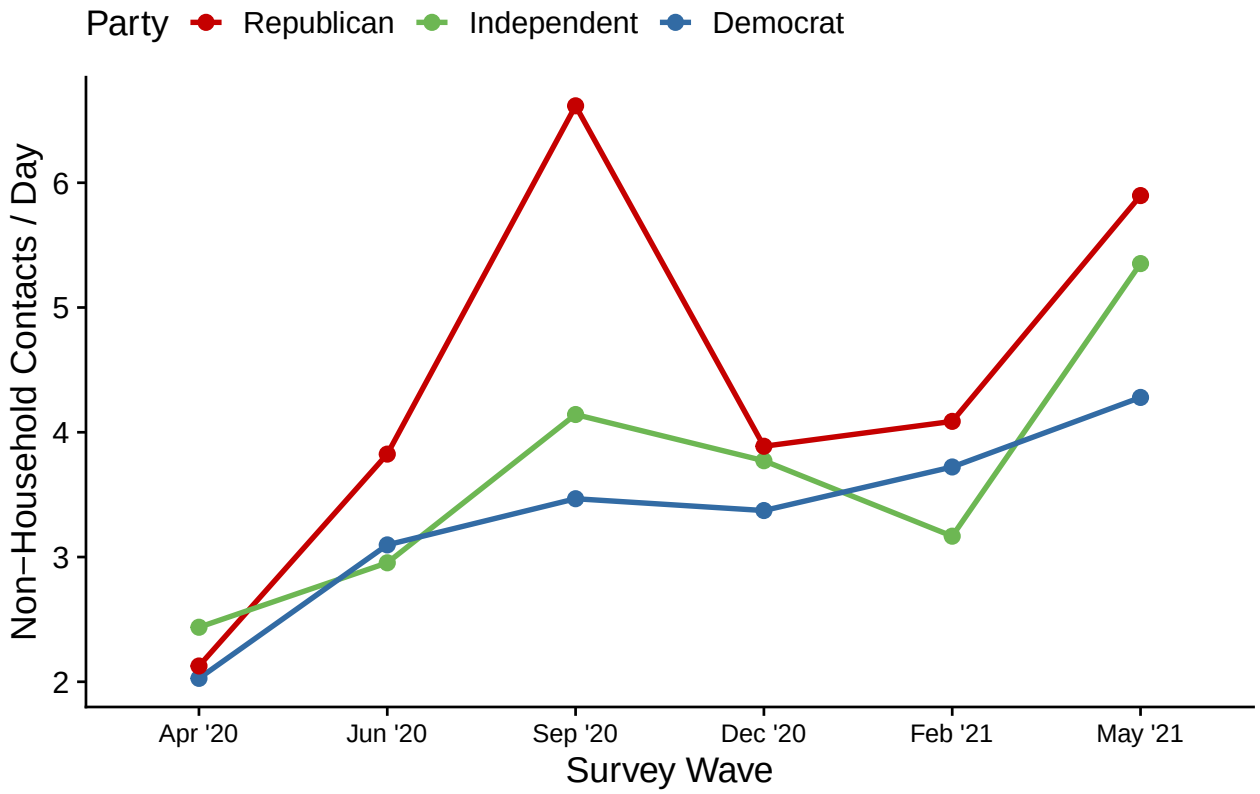

Figure 13: Non-Household Contacts Over Time by Party. Lines show weighted mean non-household (discretionary) contacts per day across survey waves. The partisan gap persists throughout the study period.

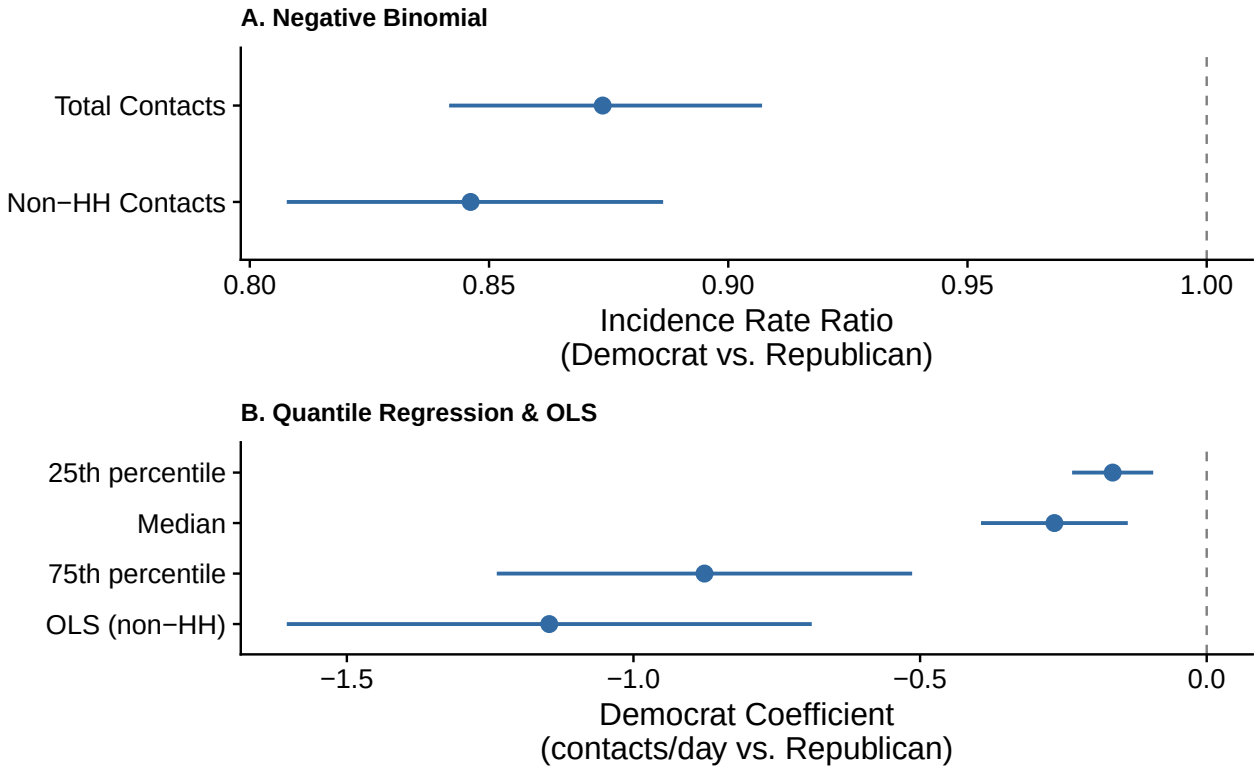

Figure 14: Partisan Contact Gap Across Model Specifications. Points show the estimated Republican coefficient (relative to Democrat baseline) with 95% confidence intervals. Negative binomial estimates are incidence rate ratios (IRR); quantile regression and OLS estimates are in raw contact units. All models control for survey wave, age, race/ethnicity, gender, education, household size, local political context, and local COVID-19 mortality.

Across all specifications, Democrats report significantly fewer contacts than Republicans. The negative binomial models (Panel A) show that Democrats have a lower rate of both total and non-household contacts relative to Republicans, confirming that the gap is not an artifact of the OLS functional form. The quantile regressions (Panel B) reveal that the Democrat–Republican gap is present not only at the median but also at the 25th and 75th percentiles of the contact distribution, ruling out the possibility that the mean difference is driven by a small number of high-contact Republican outliers. The OLS estimate for non-household contacts corroborates the negative binomial finding. Together, these results demonstrate that the partisan contact gap is (a) concentrated in discretionary non-household contacts, (b) robust to modeling overdispersed count data, and (c) present across the full distribution of contacts—not merely an artifact of the mean.

6.6.1 Restricted Sample: Respondents with 2+ Non-Household Contacts

A final concern is that partisan gaps might be inflated by respondents reporting very few non-household contacts, where small absolute differences translate to large proportional differences and where mask-usage is noisier (masks are less relevant when contacts occur primarily within the household). To address this, we restrict the sample to respondents reporting 2 or more non-household contacts (N = 9,444).

Figure 15 shows non-household contacts by party over time in this restricted sample. The partisan gap remains clearly visible: Republicans report 1.43 more non-household contacts per day than Democrats on average, and the OLS coefficient for Democrat (vs. Republican) is -1.28 ($p < .001$) after controlling for wave, demographics, and local conditions.

Figure 16 shows mask usage in the same restricted sample. Democrats report 11.1 percentage

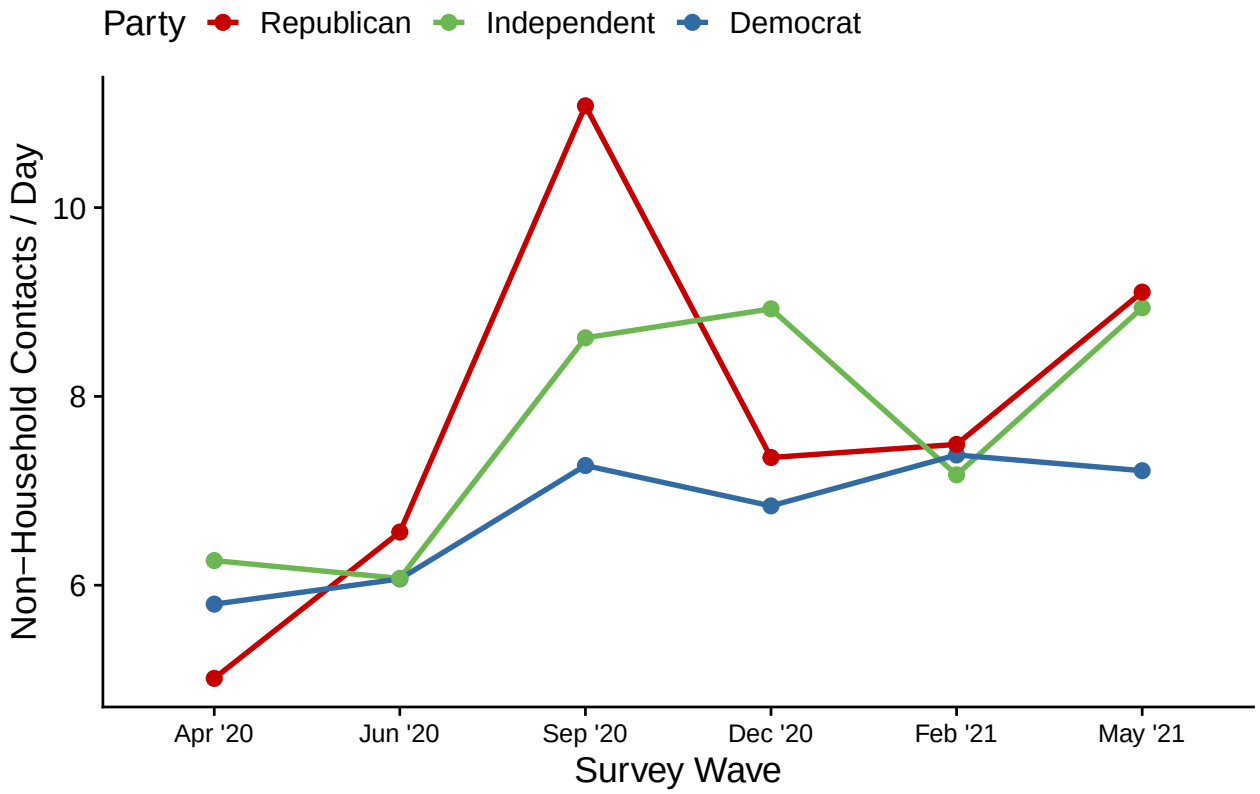

Figure 15: Non-Household Contacts Over Time by Party, Restricted to Respondents with 2+ Non-Household Contacts. Lines show weighted mean non-household contacts per day. The partisan gap persists after excluding low-contact respondents.

points higher mask usage than Republicans on average, and the OLS coefficient for Democrat (vs. Republican) is 9.0 pp ($p < .001$) after the same controls.

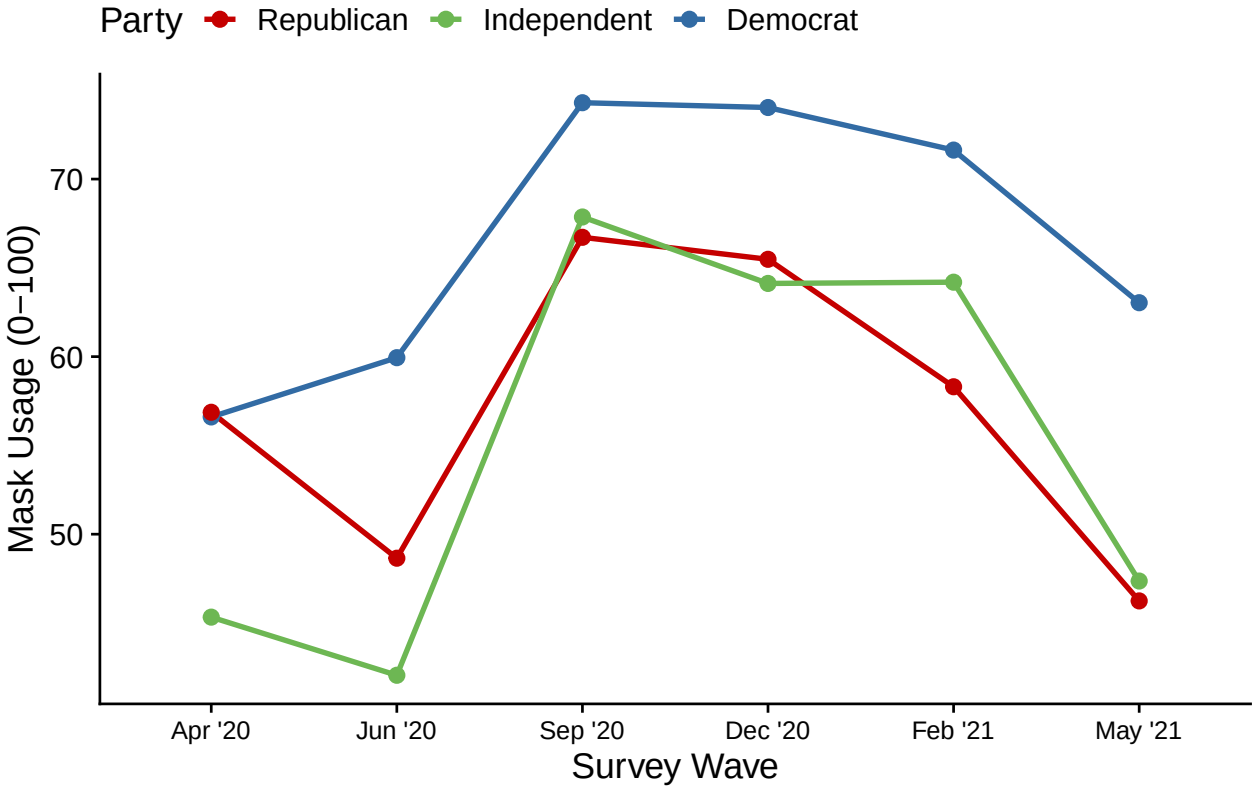

Figure 16: Mask Usage Over Time by Party, Restricted to Respondents with 2+ Non-Household Contacts. Lines show weighted mean mask usage (0–100 scale) across survey waves. The partisan gap persists among respondents with meaningful non-household contact activity.

These results confirm that partisan gaps in both contact behavior and mask usage are not driven by respondents with minimal non-household contact activity. Among those with 2 or more discretionary contacts—respondents for whom both contact reduction and masking decisions are most consequential—Republicans still report more contacts and Democrats still report substantially higher mask usage.

6.7 APPENDIX G. Pandemic Risk Perceptions Over Time

The main text documents substantial partisan differences in pandemic risk perception, with Democrats more likely than Republicans to report being "very concerned" about COVID-19. Figure 17 shows the overall distribution of concern by party, pooled across all waves. This section then examines how these perception gaps evolved over the course of the pandemic, paralleling our analysis of behavioral gaps over time.

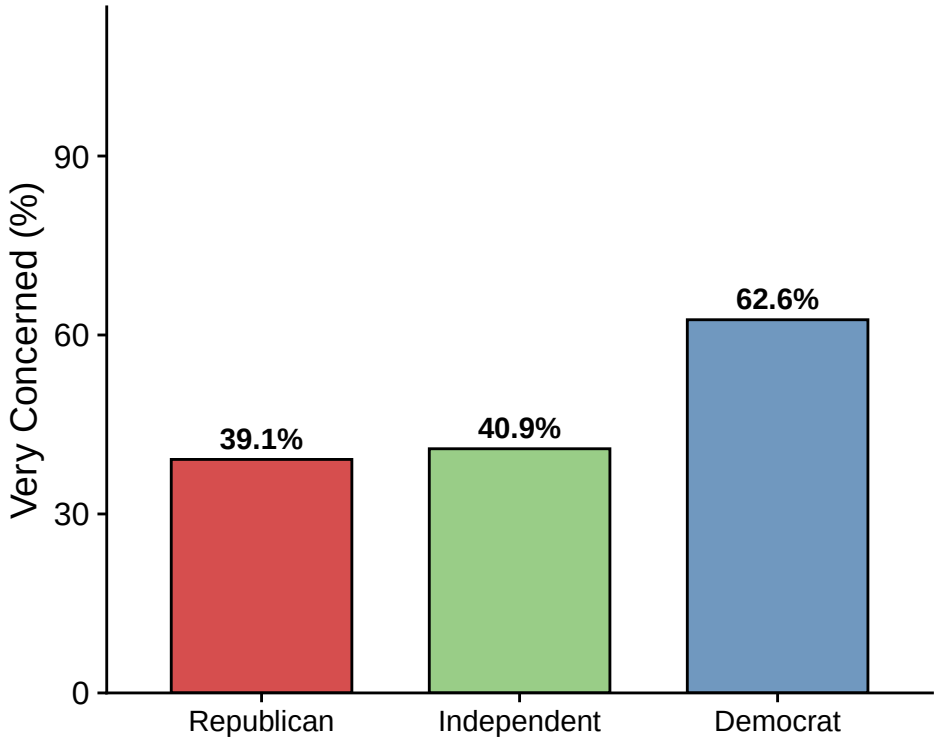

Figure 17: Partisan Differences in Pandemic Risk Perception. Bars show the percentage of respondents reporting being 'very concerned' about COVID-19, by political party. Values are weighted means using raked survey weights, pooled across all survey waves.

Figure 18 displays the percentage of each partisan group reporting being "very concerned" about COVID-19 across survey waves. Figure 19 shows the Democrat-Republican gap in this measure over time, with weekly COVID-19 cases shown in the background for context.

The perception gap was smallest at the beginning of the pandemic: just 11.3 percentage points in

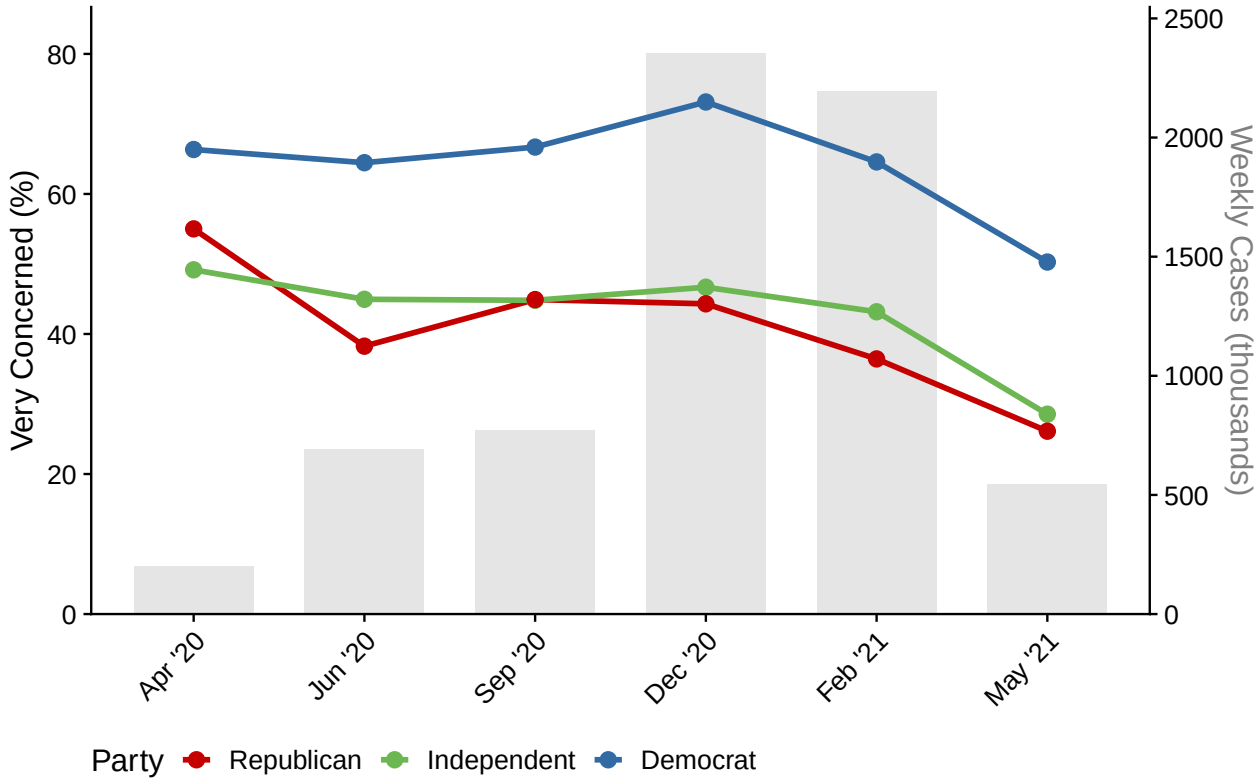

Figure 18: Pandemic Risk Perception by Party Across Survey Waves. Lines show the percentage of respondents reporting being 'very concerned' about COVID-19. Gray background bars indicate weekly COVID-19 cases (in thousands).

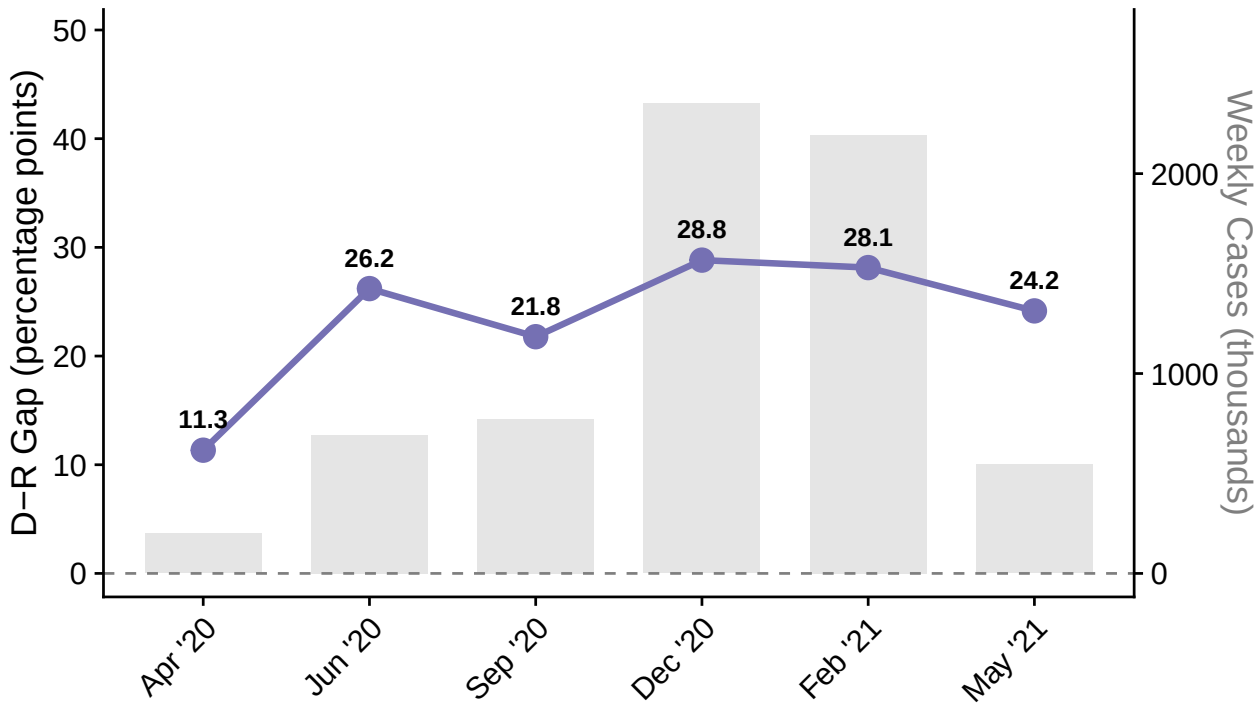

Figure 19: Partisan Gap in Pandemic Risk Perception Across Survey Waves. Points show the Democrat-Republican difference in the percentage reporting being 'very concerned' about COVID-19. Gray background bars indicate weekly COVID-19 cases (in thousands).

Apr '20. By Dec '20, the gap had grown to 28.8 percentage points. In April 2020, Republicans were nearly as likely as Democrats to report being "very concerned" about COVID-19; by June 2020, that gap had more than doubled, and it remained elevated throughout the remainder of the study period.

Critically, while incidence was relatively low in April 2020, *mortality rates were at their highest*—the virus was simply more deadly in the early months before treatments improved and clinical experience accumulated. This pattern is consistent with our broader finding that concrete, visible consequences reduce polarization. When the threat was most lethal, perceptions were least polarized. As case fatality rates declined over subsequent months—even as case counts rose—the perception gap widened. The combination of declining lethality and escalating politicization of the pandemic through debates over lockdowns, masks, and the credibility of public health guidance

likely both contributed to this divergence.

All three partisan groups showed declining concern over time, particularly from the winter surge through spring 2021 as vaccines became widely available. However, this decline was steeper among Republicans and Independents than among Democrats, maintaining the perception gap even as overall concern fell.

6.7.1 Perceptions vs. Local Pandemic Conditions

The previous figures show perceptions against *national* case counts. But Republicans and Democrats live in different places and thus experienced different local pandemic conditions. Figure 20 shows each party's stated concern alongside the average local incidence rate they actually experienced, given where respondents in each party lived. This allows us to assess whether perceptions tracked the objective conditions each group faced.

The figure reveals that Republicans and Democrats experienced similar local incidence trajectories—both groups saw the winter 2020–21 surge—yet their perceptions diverged over time. Notably, incidence was relatively low in April 2020 when the perception gap was smallest, complicating any simple story about incidence driving perceptions.

Figure 21 shows the same comparison using local mortality rates rather than incidence. Mortality is a more concrete and visible indicator of pandemic severity—deaths are reported in local news, experienced through social networks, and marked by obituaries—making it a stronger test of whether perceptions tracked objective conditions.

The mortality figure reveals a striking pattern: local mortality rates were actually *highest* in April 2020—precisely when the perception gap was smallest. The virus was simply more deadly in

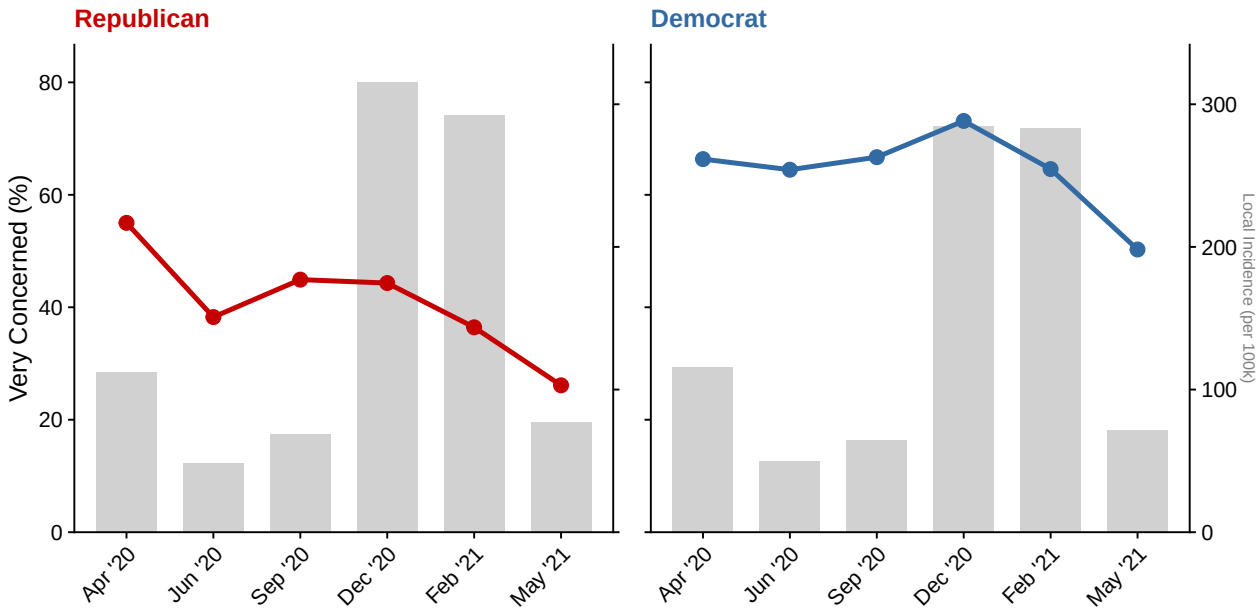

Figure 20: Perceptions vs. Local Pandemic Conditions by Party. Each panel shows one party's stated concern (line) against the average local incidence rate experienced by that party's respondents (bars). Local incidence is calculated as the weighted average of county-level weekly incidence rates among respondents of each party.

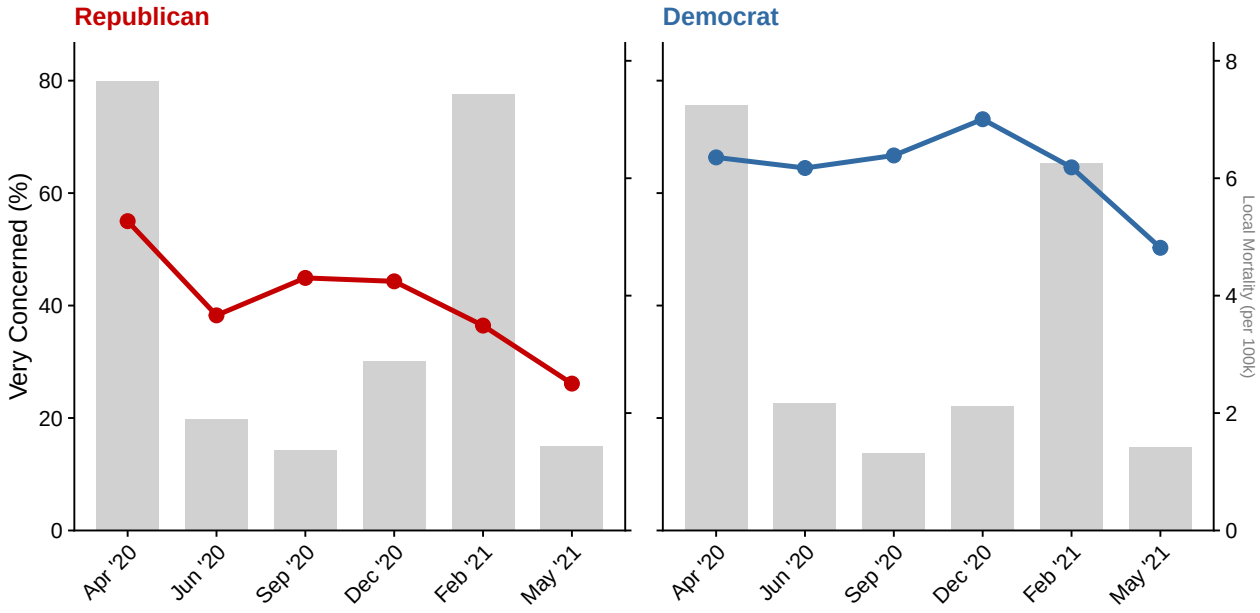

Figure 21: Perceptions vs. Local Mortality by Party. Each panel shows one party's stated concern (line) against the average local mortality rate experienced by that party's respondents (bars). Local mortality is calculated as the weighted average of county-level weekly mortality rates (per 100,000) among respondents of each party.

the early months before treatments improved and clinical protocols developed. As mortality rates declined over subsequent months, the perception gap widened. This pattern supports our broader thesis that concrete, visible consequences reduce polarization: when the threat was most lethal, partisans were most aligned in their concern.

The winter 2020–21 surge presents a more complex picture. Mortality rose sharply, and Republicans did reduce their contacts in response—yet their stated concern continued to decline. This disconnect between mortality-driven *behavior* and diverging *perceptions* suggests that by this point, expressed concern had become entangled with partisan identity in ways that were resistant to objective conditions. The early-pandemic alignment, when partisanship had not yet fully politicized COVID-19, may represent a window before symbolic boundaries hardened. Once the pandemic became a site of partisan conflict, stated perceptions became partly performative—reflecting group allegiance rather than risk assessment—even as underlying behavior continued to respond to concrete threat.

6.7.2 Health Behaviors by Concern Level

Figure 22 shows health behaviors by party within each level of stated concern. This complements Figure 3 in the main text by showing the absolute levels rather than the gaps.

Figure 23 shows how behavior among the "very concerned" subgroup evolved over time, alongside the changing share of each party reporting high concern. This analysis reveals that the meaning of "very concerned" changed over the pandemic as different types of partisans selected into this category.

Panel A shows that the share reporting "very concerned" declined for both parties, but especially

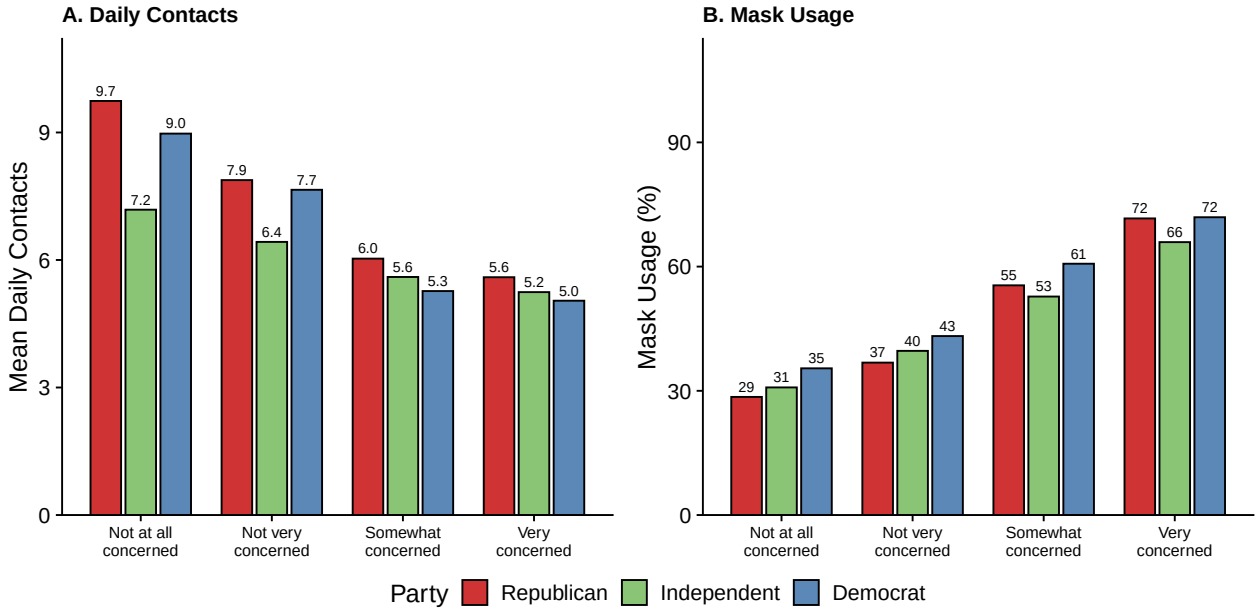

Figure 22: Health Behaviors by Party Within Each Level of Stated Concern. Panel A shows mean daily contacts; Panel B shows mask usage. Sample sizes: Not at all concerned (n=1,127), Not very concerned (n=1,897), Somewhat concerned (n=5,894), Very concerned (n=9,168).

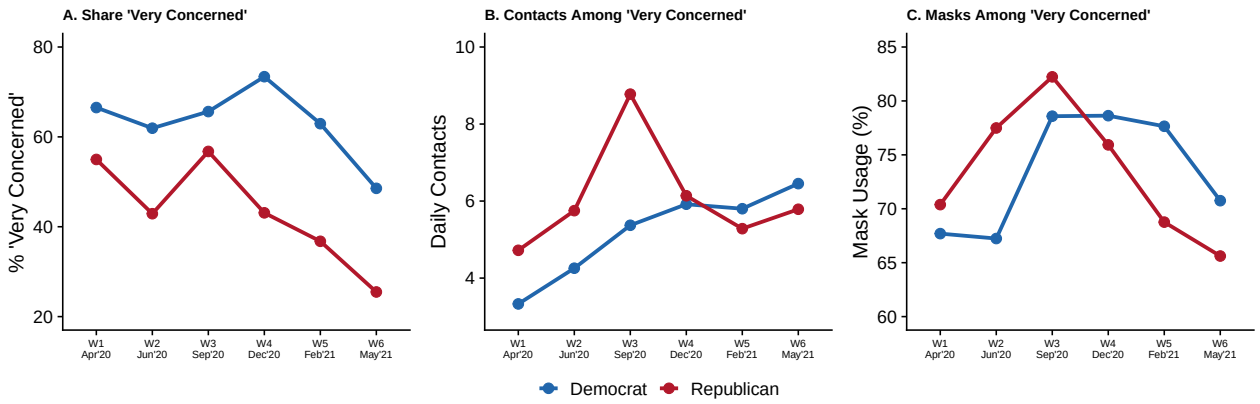

Figure 23: Behavior Among the 'Very Concerned' Over Time. Panel A shows the percentage reporting being 'very concerned'—declining for both parties but especially Republicans (55% to 26%). Panels B and C show behavior *only among those very concerned*. The contact gap (Panel B) was positive early on but collapsed during the winter surge and reversed by wave 5-6. The mask gap (Panel C) reversed earlier (wave 4) and persisted, suggesting politicization independent of concern level.

for Republicans (55% to 26% vs. 67% to 49% for Democrats). Panels B and C reveal how behavior among the very concerned evolved—and the patterns differ in revealing ways.

For contacts (Panel B), very concerned Republicans initially reported *more* contacts than very concerned Democrats (gap of +1.4 to +3.4 contacts in waves 1-3). But this gap collapsed during the winter 2020-21 surge (wave 4-5) and actually reversed by waves 5-6, with concerned Democrats reporting slightly more contacts. This timing aligns with the lifting of stay-in-place restrictions: among all respondents, the percentage reporting being under a stay-in-place order declined from approximately 95% in wave 1 to 70-81% by wave 5, with Republicans showing a larger decline than Democrats. The contact pattern suggests mortality-driven convergence early on, followed by Democrats increasing contacts as restrictions lifted—perhaps reflecting a strategic shift toward relying on masks and vaccines rather than contact reduction.

For masks (Panel C), a similar reversal occurred but with different timing: very concerned Republicans wore masks at *higher* rates than very concerned Democrats in waves 1-3, but by wave 4 Democrats wore masks more. Unlike contacts, this reversal did not correspond to the mortality surge or the lifting of restrictions—it persisted even as mortality declined in waves 5-6. This pattern suggests that mask-wearing became increasingly tied to partisan identity over time, such that even "very concerned" Republicans reduced masking as the behavior became symbolically marked.

6.7.3 Wave 1 Only: Gaps by Concern Level When Mortality Was Highest

The pooled analysis above combines data from all waves. To examine whether the relationship between concern and partisan gaps differed when objective threat was highest, Figure 24 replicates this analysis for Wave 1 (April 2020) only—the period when mortality rates were at their peak.

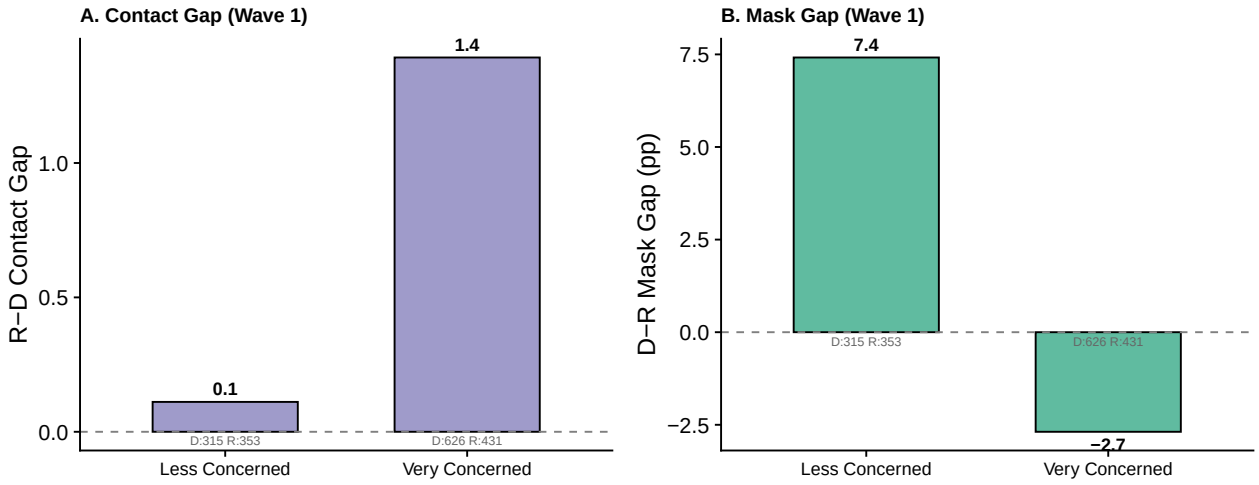

Figure 24: Partisan Gaps in Health Behaviors by Concern Level, Wave 1 Only (April 2020). Panel A shows the Republican-Democrat gap in daily contacts; Panel B shows the Democrat-Republican gap in mask usage. Wave 1 represents the period of highest mortality rates, before the pandemic became heavily politicized. Binary concern grouping (Very Concerned vs. Less Concerned) used for statistical power. Unweighted means used since survey weights were designed for the full sample. Sample sizes shown on each bar (D = Democrats, R = Republicans).

Comparing Wave 1 to the pooled results reveals whether the early pandemic—before extensive politicization—showed different patterns. If concrete mortality threat reduces polarization, we might expect smaller or even reversed gaps in Wave 1 compared to later periods.

6.7.4 The Politicization of Mask-Wearing: Exact Gap Values

Figure 23 above shows how mask-wearing patterns among the "very concerned" reversed over time, alongside the changing composition of who reports being very concerned. Figure 25 provides the exact D-R gap values for each wave.

As discussed in the main text, two dynamics likely contribute to this reversal: (1) masks became increasingly politicized as partisan signals, and (2) the Republicans who continued to report being "very concerned" became an increasingly select and unusual subset as overall Republican concern declined.

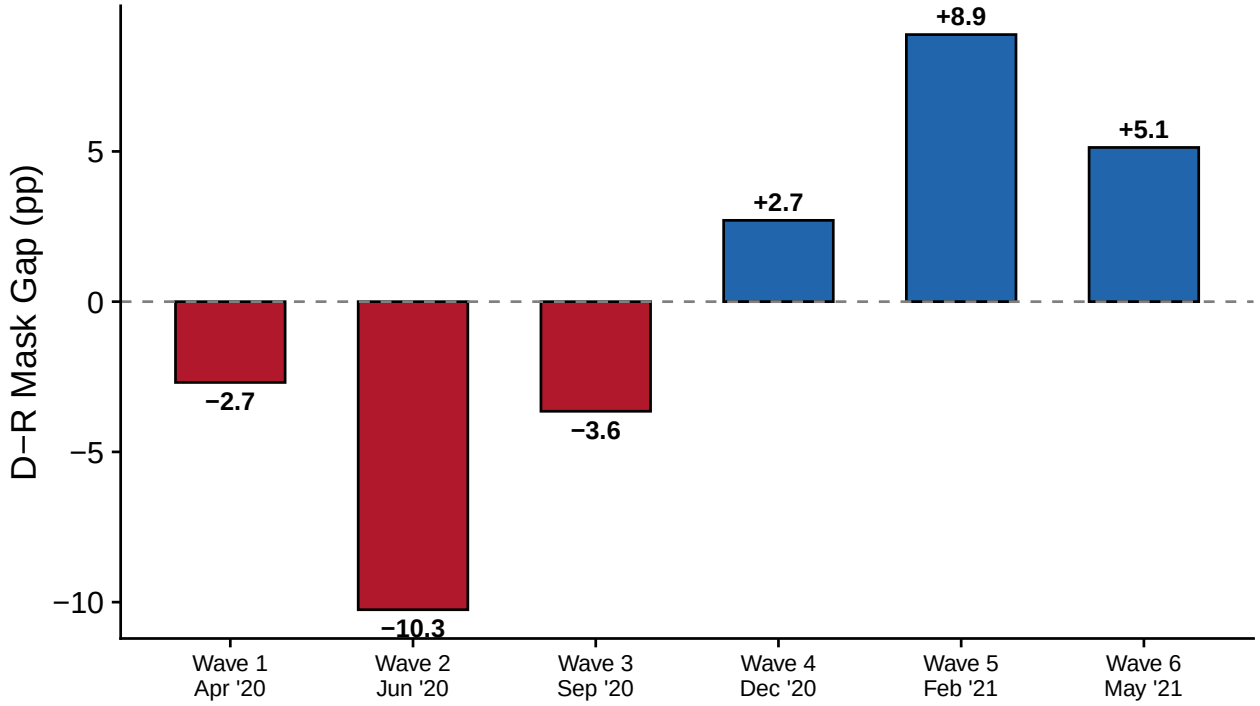

Figure 25: D-R Mask Gap Among 'Very Concerned' Respondents Over Time. Bars show the Democrat-Republican difference in mask usage among those reporting being 'very concerned' about COVID-19. Negative values (red) indicate Republicans wore masks more; positive values (blue) indicate Democrats wore masks more. The reversal from waves 1-3 to waves 4-6 is evident. See Figure 23 for context on changing composition.

6.8 APPENDIX H. Detailed Pandemic Response by Party

Figure 26 shows the standardized regression coefficients for the relationship between prior week's pandemic conditions (incidence and mortality) and health behaviors (contacts and mask usage), by party. Effects are standardized: bars show the change associated with a one standard deviation increase in log incidence or log mortality.

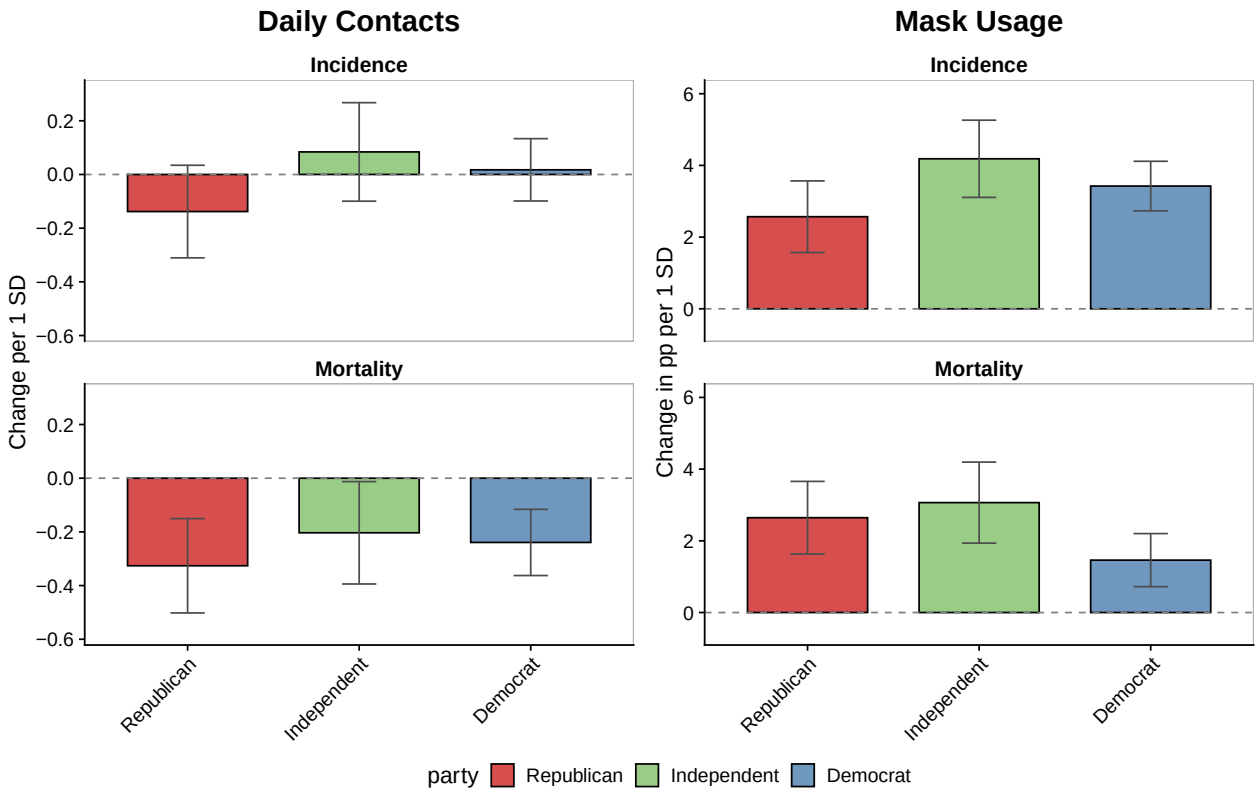

Figure 26: Detailed Behavioral Response to Prior Week's Pandemic Conditions by Party. Left column shows changes in daily contacts; right column shows changes in mask usage (percentage points). Top row shows response to incidence; bottom row shows response to mortality. Effects are standardized: bars show the change associated with a one standard deviation increase in log incidence or log mortality. Models include county fixed effects. Error bars indicate standard errors. Negative values for contacts indicate reductions; positive values for masks indicate increases.

Figure 27 summarizes how partisan gaps change in response to pandemic conditions. This figure complements the individual-level effects shown above by directly displaying the implications for polarization.

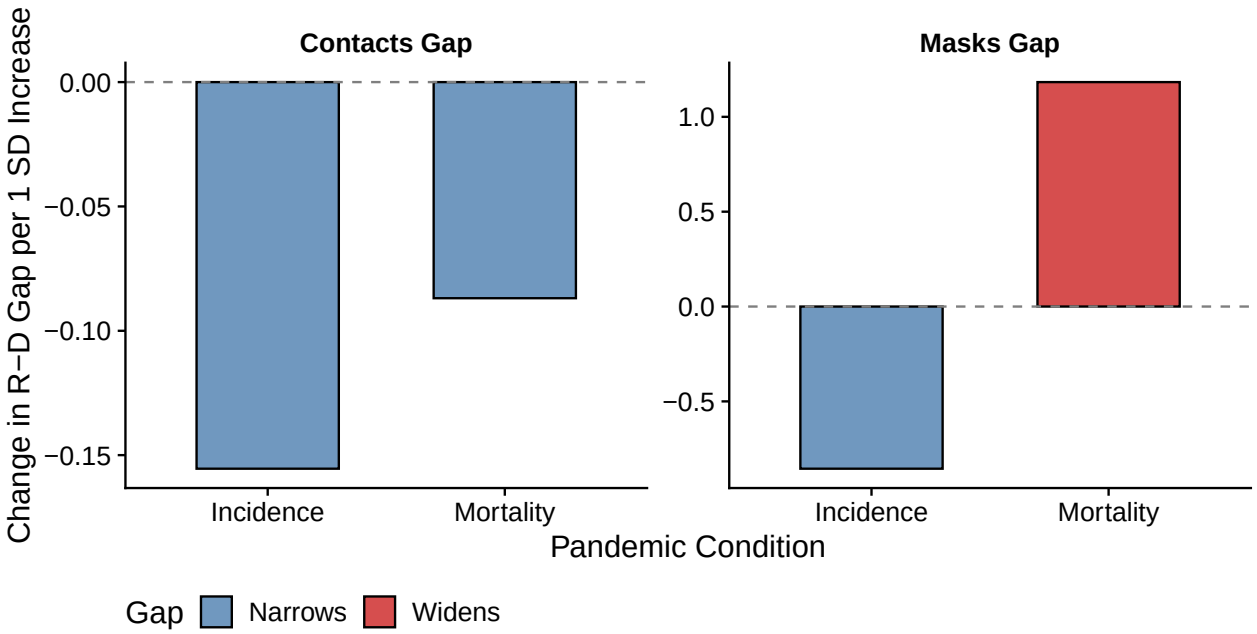

Figure 27: Effect of Prior Week's Pandemic Conditions on Partisan Gaps. Bars show how the Republican-Democrat gap changes in the week following a one standard deviation increase in local log incidence or mortality. Negative values indicate gap narrowing (convergence); positive values indicate gap widening (divergence).

6.9 APPENDIX I. Spatial Variation in Partisan Differences

This section provides additional detail on geographic variation in partisan differences. All estimates in this section are adjusted for survey wave and local mortality rates using regression models, removing potential confounding from differences in when surveys were collected and local pandemic severity. We present behavior levels by party within census regions (complementing the region-level gap maps in the main text), then examine finer geographic scales: census divisions (9 divisions) and states.

?@fig-region-levels-map displays adjusted health behavior levels by party and census region.

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6.9.1 Partisan Gaps Within Policy Environments

A potential confounder for geographic patterns in partisan gaps is that metropolitan areas were more likely to have strict mask mandates. Figure 28 shows that partisan gaps persist within each policy environment, indicating that individual partisanship shapes behavior beyond policy effects. The mask mandate analysis is restricted to waves 1-4 (April-December 2020), when mandates were most actively in force; by waves 5-6, many jurisdictions had begun relaxing mandates as vaccination progressed.

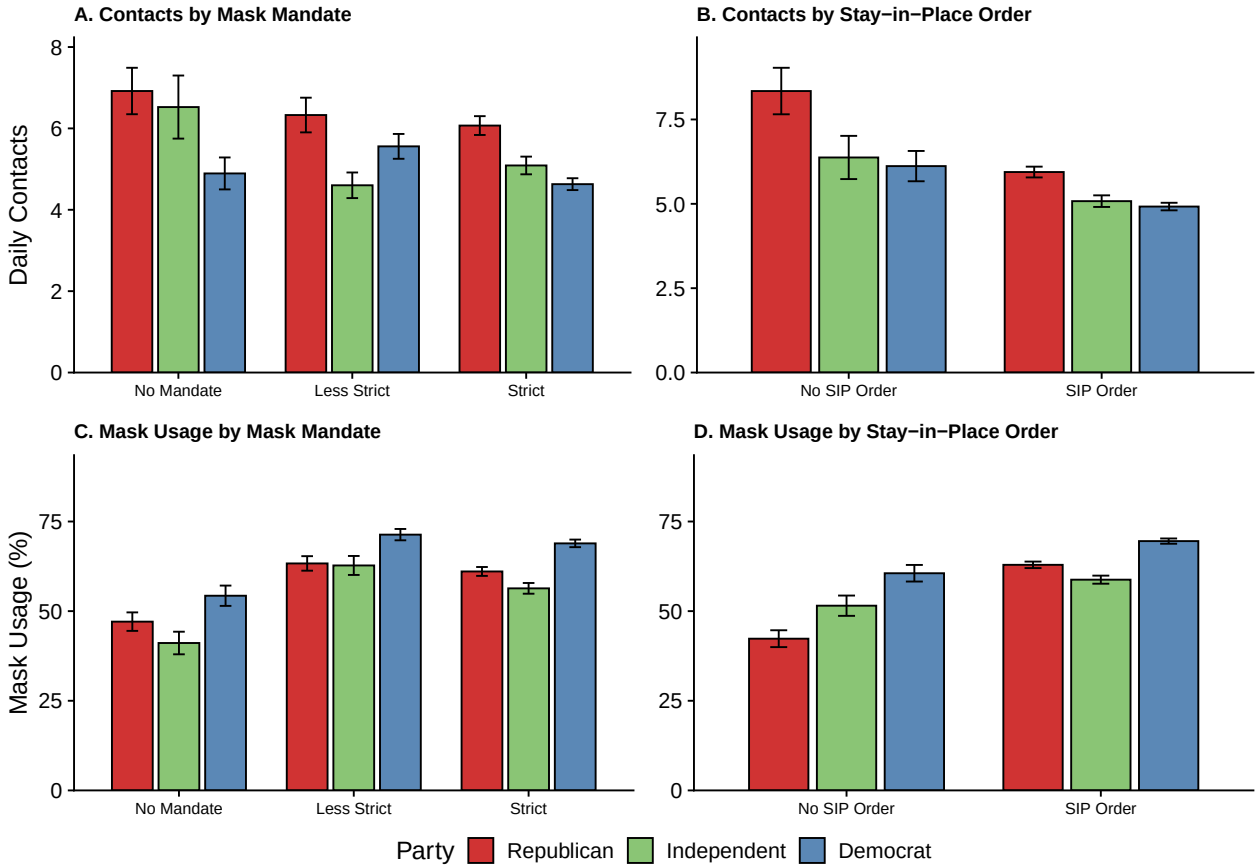

Figure 28: Health Behaviors by Party and Policy Environment. Left column shows behavior by county mask mandate policy (no mandate, less strict, strict; waves 1-4 only, April-December 2020). Right column shows behavior by respondent-reported stay-in-place order status (waves 1-5 only; this question was not assessed in wave 6). Top row shows daily contacts; bottom row shows mask usage. Partisan gaps persist within each policy environment.

6.9.2 Behavior by Local Partisan Context

Figure 29 shows daily contacts by party and local partisan context (congressional district representation). The figure reveals that partisan gaps in contacts persist regardless of local context, though the magnitude varies.

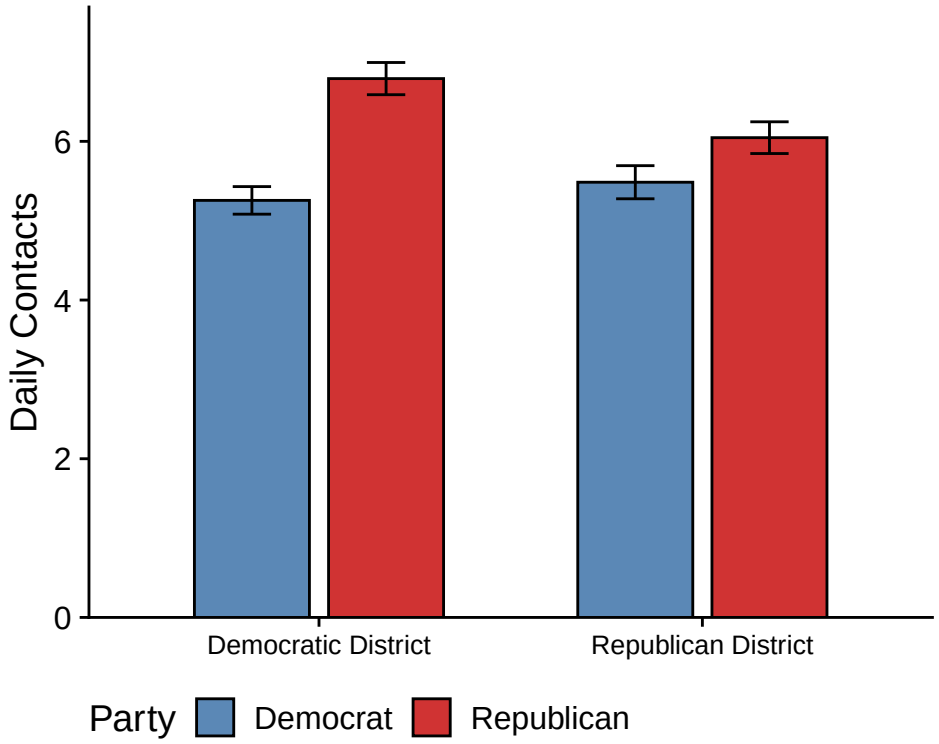

Figure 29: Daily Contacts by Party and Local Partisan Context. Bars show mean daily contacts for Democrats and Republicans depending on whether they live in a Democratic-led or Republican-led congressional district. Estimates are adjusted for survey wave, local incidence rates, and metropolitan status. Error bars indicate standard errors.

Figure 30 shows the Republican-Democrat gap in each behavior by local partisan context. For contacts, the R-D gap is larger in Democratic-led districts than in Republican-led districts, suggesting that Republicans increase contacts more than Democrats when living in Democratic areas. For masks, the D-R gap is similar across contexts, indicating that both parties shift their mask behavior in parallel when living in opposing-party areas.

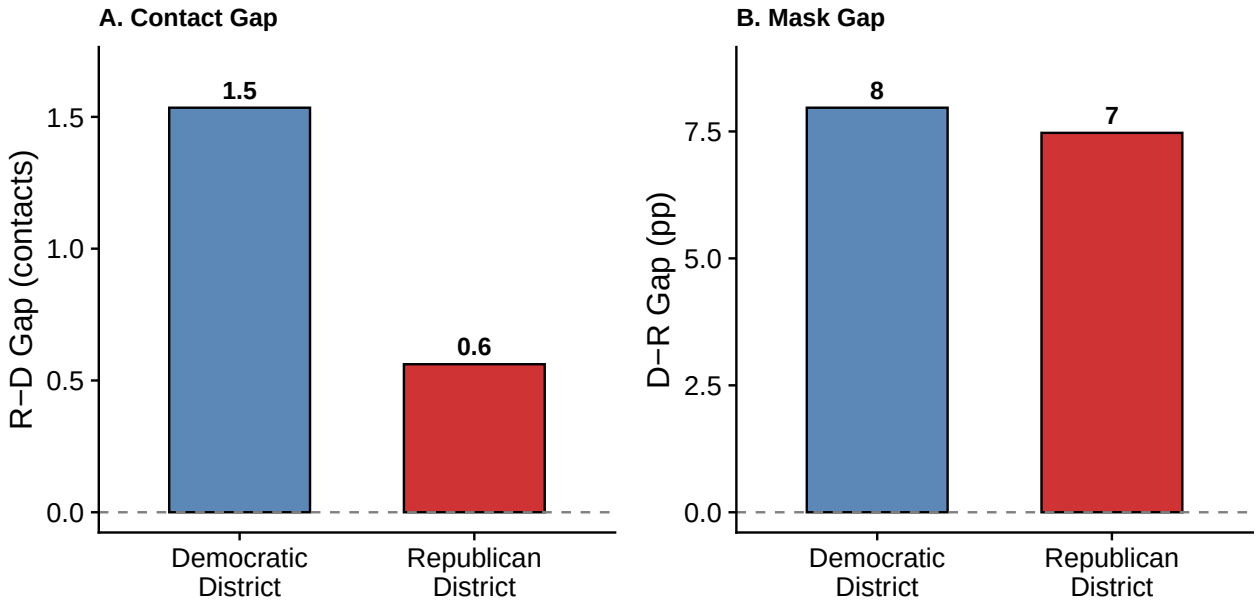

Figure 30: Partisan Gaps by Local Context. Panel A shows the Republican-Democrat gap in daily contacts; Panel B shows the Democrat-Republican gap in mask usage. Gaps are calculated within each congressional district context (Democratic-led vs Republican-led). Estimates are adjusted for survey wave, local incidence rates, and metropolitan status.

6.9.3 Regional Behavior Levels

Figure 31 displays the raw behavior levels (mean contacts and mask usage) for each party within each census region. This complements the gap-focused maps in the main text by showing the underlying distributions.

The figure reveals that the smaller partisan gaps in the South are driven primarily by Republicans in that region reporting fewer contacts and higher mask usage than Republicans elsewhere, rather than by Democrats in the South behaving differently. This suggests that regional norms may exert a moderating influence on Republican behavior specifically.

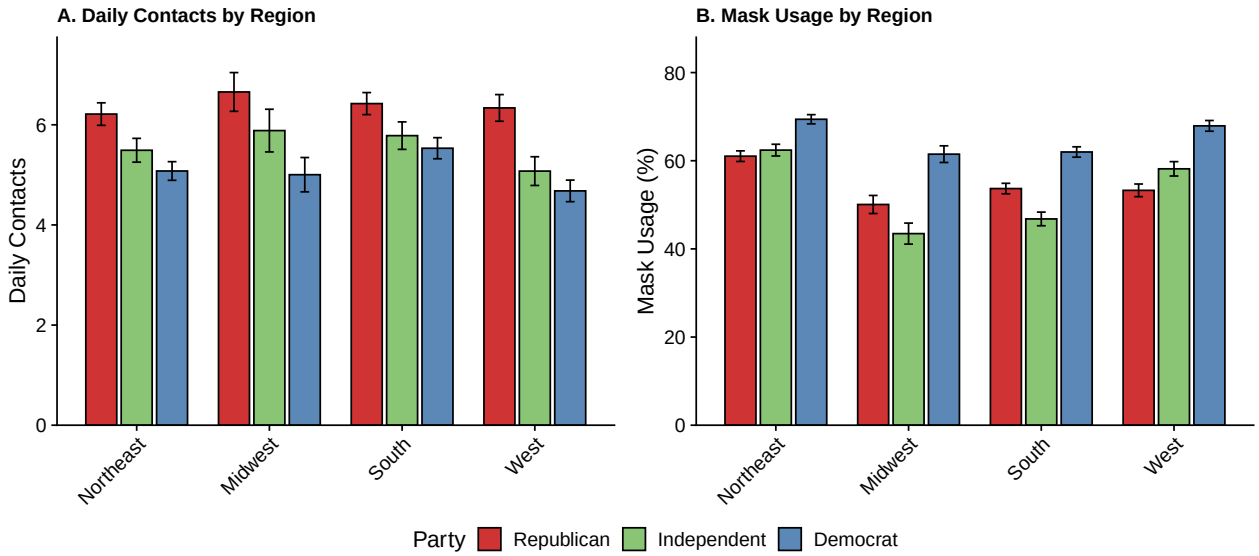

Figure 31: Adjusted Health Behaviors by Party Across Census Regions. Panel A shows mean daily contacts; Panel B shows mask usage (percentage of contacts with masks). Estimates are adjusted for survey wave and local mortality rates. Error bars indicate standard errors from emmeans.

6.9.4 Regional Vaccine Acceptance and Perception Maps by Party

Figure 32 displays vaccine acceptance rates (Wave 6 only) and pandemic concern levels by party across census regions. These complement the contact and mask usage maps above.

Vaccine acceptance rates show strong partisan gaps across all regions, with Democrats consistently reporting higher acceptance rates than Republicans. The gap is largest in the South and Midwest. Perception of pandemic severity (concern) shows an even starker partisan divide: Democrats report being "very concerned" about COVID-19 at rates 20-30 percentage points higher than Republicans in every region. Interestingly, while Republican concern varies less across regions, Democratic concern is highest in the Northeast and West.

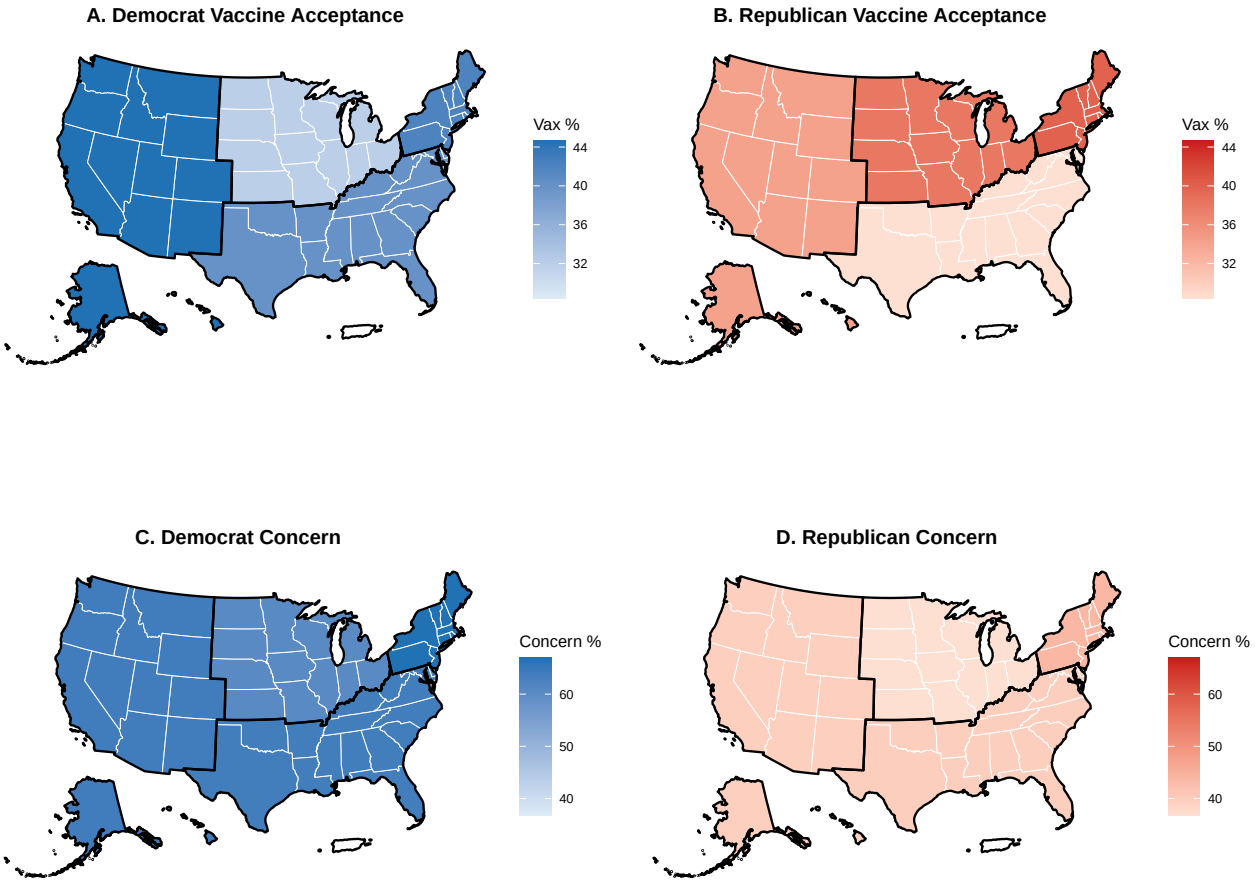

Figure 32: Adjusted Vaccine Acceptance and Perception Levels by Party Across Census Regions. Top row shows vaccine acceptance rates (Wave 6, May 2021) for Democrats (A) and Republicans (B). Bottom row shows percentage reporting being 'very concerned' about COVID-19 for Democrats (C) and Republicans (D). Estimates are adjusted for local mortality rates. Darker shading indicates higher values. Black borders indicate census region boundaries.

6.9.5 All Partisan Gaps by Census Region

Figure 33 displays partisan gaps across all four outcomes in a single 2×2 view, facilitating comparison of geographic patterns across behaviors and perceptions.

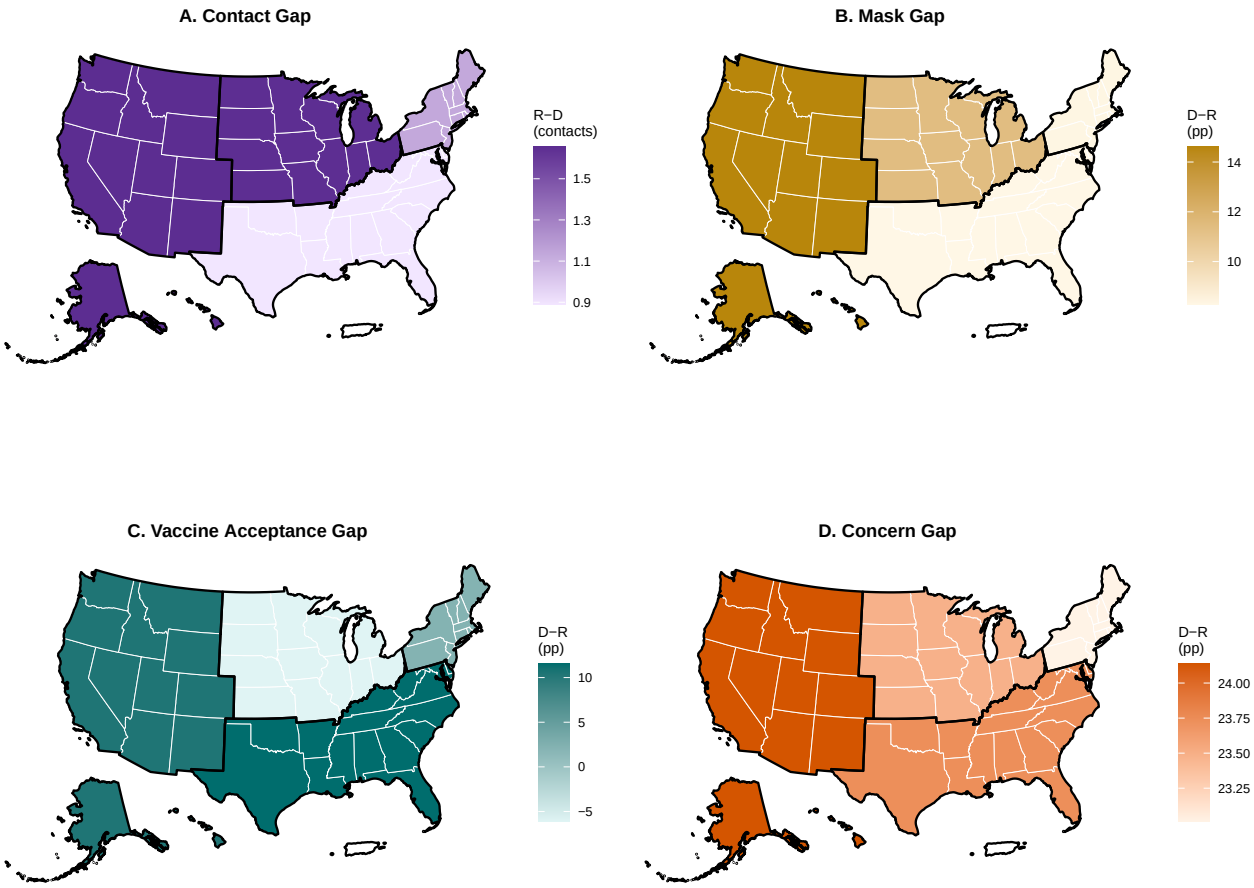

Figure 33: Geographic Distribution of All Partisan Gaps by Census Region (Adjusted). Panel A shows the Republican-Democrat gap in daily contacts (purple); Panel B shows the Democrat-Republican gap in mask usage (gold); Panel C shows the Democrat-Republican gap in vaccine acceptance (teal); Panel D shows the Democrat-Republican gap in COVID-19 concern (orange). Estimates are adjusted for survey wave and local mortality rates. Black borders indicate census region boundaries.

The 2×2 view reveals that geographic patterns differ across outcomes. Contact gaps (A) and mask gaps (B) show the most regional variation, with the South having notably smaller gaps. In contrast, vaccine acceptance gaps (C) and especially concern gaps (D) are more uniform across regions, suggesting that these outcomes are more strongly driven by partisanship itself rather than regional

context.

6.9.6 City-Level Partisan Gaps

Figure 34 displays partisan gaps in contacts and mask usage across the six BICS sample cities plus the national average. These city-level comparisons complement the regional analysis by showing that partisan gaps persist even within individual metropolitan areas.

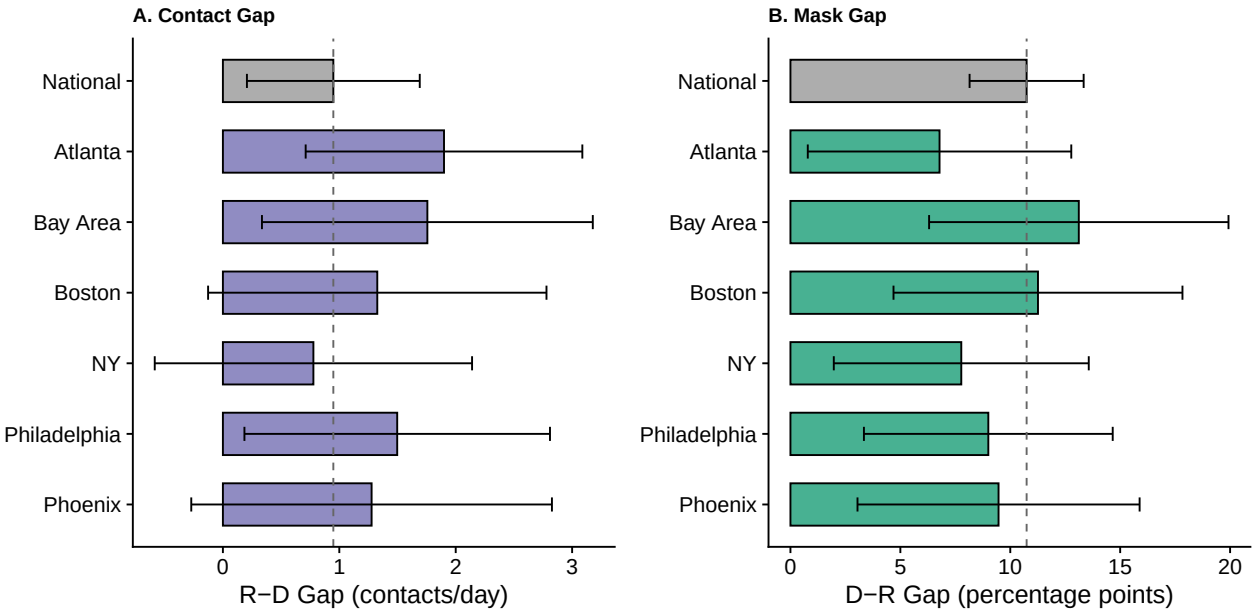

Figure 34: Partisan Gaps in Health Behaviors by City. Panel A shows the Republican-Democrat gap in daily contacts; Panel B shows the Democrat-Republican gap in mask usage (percentage points). Error bars show 95% confidence intervals based on pooled standard errors. Dashed line indicates national average gap.

Partisan gaps appear in every city, though their magnitude varies. The contact gap is positive (Republicans reporting more contacts) in all cities, and the mask gap is positive (Democrats masking more) in all cities. This within-city persistence reinforces the main text finding that partisan behavioral differences are not artifacts of geographic sorting.

6.10 APPENDIX J. Partisan Behavior Levels Over Time

The main text documents how partisan gaps in health behaviors varied over the course of the pandemic. To understand what drives these fluctuations, Figure 35 displays the average behavior levels for each party across survey waves. This decomposition reveals an important asymmetry: **Democrats maintained relatively stable behavior throughout the pandemic, while Republicans showed substantially greater variability**.

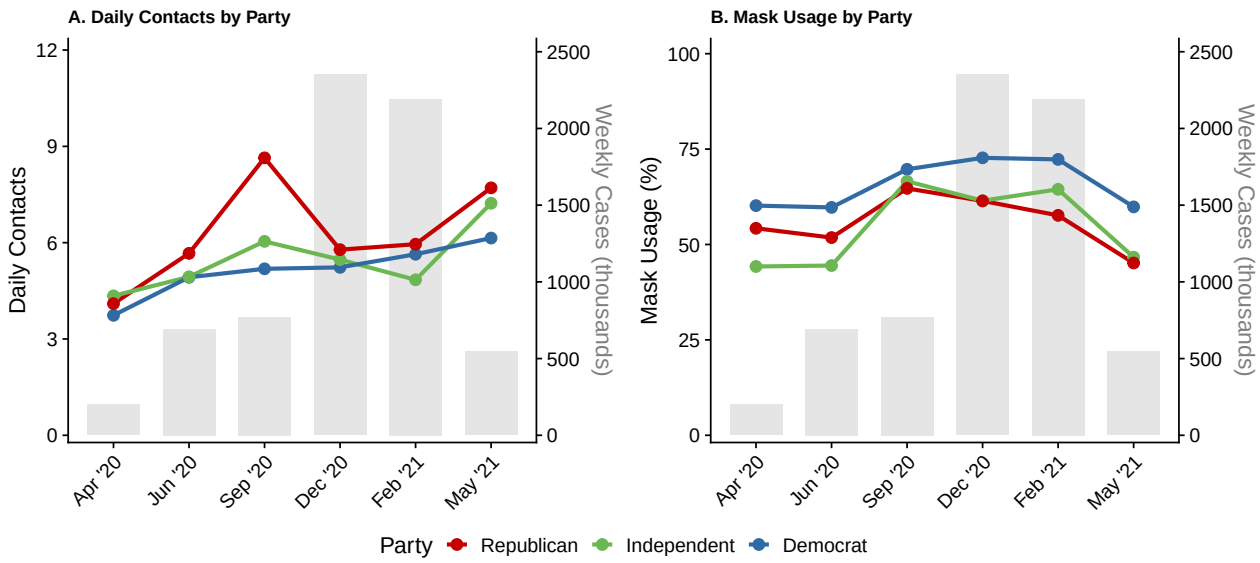

Figure 35: Health Behavior Levels by Party Across Survey Waves. Panel A shows mean daily contacts; Panel B shows mask usage (percentage of contacts with masks). Gray background bars indicate weekly COVID-19 cases (in thousands). Democrats (blue) maintained relatively stable behavior while Republicans (red) showed greater variability, particularly in contact behavior.

For contact behavior (Panel A), Democrats maintained a relatively narrow range of daily contacts across all six waves, while Republicans exhibited pronounced swings—increasing contacts substantially during periods of lower case counts (September 2020) and reducing them during surges (December 2020, February 2021). The coefficient of variation for Republican contacts (25.7%) was notably higher than for Democrats (15.8%), confirming greater Republican variability.

Mask usage (Panel B) shows a different pattern. All parties increased mask-wearing from the early

pandemic through winter 2020–21, but Republicans started from a lower baseline and showed steeper declines as vaccines became available in spring 2021. Democrats maintained consistently high mask usage throughout, while Republican mask-wearing tracked more closely with pandemic conditions. This asymmetry—stable Democratic behavior paired with variable Republican behavior—is the primary driver of the fluctuating partisan gaps documented in the main text.

6.10.1 Formal Test of Variance Differences

To formally test whether variance in contact behavior differs across partisan groups, we conducted Levene's test for equality of variances on individual-level contact data. The results confirm significant heterogeneity in variance across parties ($F(2, 18{,}230) = 15.47$, $p < 0.001$). Republicans exhibited the highest variance in daily contacts (120.3), followed by Independents (107.9) and Democrats (83.0). Table 5 summarizes these results.

Table 5: Variance in Daily Contacts by Political Party with Levene's Test

Party	N	Mean Contacts	SD
Republican	5517	6.87384448069602	10.9667090980222
Independent	4527	5.82372432074218	10.3894697106189
Democrat	8189	5.53938209793628	9.10909661236962
Levene's Test	NA	NA	F = 15.47

Note: Levene's test assesses equality of variances across groups. Significant result ($p < 0.001$) indicates varian

The significantly higher variance among Republicans indicates that Republican contact behavior was more responsive to changing conditions—whether pandemic severity, policy environment, or

media messaging—while Democrats maintained more consistent behavior throughout the study period.